

Grade 12  
Pre-Calculus Mathematics  
Achievement Test

# **Marking Guide**

June 2025

Grade 12 pre-calculus mathematics achievement test.  
Marking Guide. June 2025

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*Disponible en français.*

While the department is committed to making its publications as accessible as possible, some parts of this document are not fully accessible at this time.

Available in alternate formats upon request.

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# General Marking Instructions

**Please do not make any marks in the student test booklets.** If the booklets have marks in them, the marks will need to be removed by departmental staff prior to sample marking should the booklet be selected.

Please ensure that

- the booklet number and the number on the *Answer/Scoring Sheet* are identical
- **students and markers use only a pencil to complete the *Answer/Scoring Sheets***
- the totals of each of the four parts are written at the bottom
- each student's final result is recorded, by booklet number, on the corresponding *Answer/Scoring Sheet*
- the *Answer/Scoring Sheet* is complete
- a photocopy has been made for school records

Once marking is completed, please forward the *Answer/Scoring Sheets* to Manitoba Education and Early Childhood Learning in the envelope provided. For more information see the administration manual.

## Marking the Test Questions

The test is composed of constructed response questions and selected response questions. Constructed response questions are worth 1 to 5 marks each, and selected response questions are worth 1 mark each. An answer key for the selected response questions can be found at the beginning of the section "Booklet 2 Questions."

To receive full marks, a student's response must be complete and correct. Where alternative answering methods are possible, the *Marking Guide* attempts to address the most common solutions. For general guidelines regarding the scoring of students' responses, see Appendix A.

## Irregularities in Provincial Tests

During the administration of provincial tests, supervising teachers may encounter irregularities. Markers may also encounter irregularities during local marking sessions. The appendix provides examples of such irregularities as well as procedures to follow to report irregularities.

If a *Answer/Scoring Sheet* is marked with "0" only (e.g., student was present but did not attempt any questions), please document this on the *Irregular Test Booklet Report*.

## **Assistance**

If, during marking, any marking issue arises that cannot be resolved locally, please call Manitoba Education and Early Childhood Learning at the earliest opportunity to advise us of the situation and seek assistance if necessary.

You must contact the Assessment Consultant responsible for this project before making any modifications to the answer keys or scoring rubrics.

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## Communication Errors

The marks allocated to questions are primarily based on the concepts and procedures associated with the learning outcomes in the curriculum. For each question, shade in the circle on the *Answer/Scoring Sheet* that represents the marks given based on the concepts and procedures. A total of these marks will provide the preliminary mark.

Errors that are not related to concepts or procedures are called “Communication Errors” (see Appendix A) and will be tracked on the *Answer/Scoring Sheet* in a separate section. There is a  $\frac{1}{2}$  mark deduction for each type of communication error committed, regardless of the number of errors per type (i.e., committing a second error for any type will not further affect a student’s mark), with a maximum deduction of 5 marks from the total test mark.

When a given response includes multiple types of communication errors, deductions are indicated in the order in which the errors occur in the response. No communication errors are recorded for work that has not been awarded marks. The total deduction may not exceed the marks awarded.

The student’s final mark is determined by subtracting the communication errors from the preliminary mark.

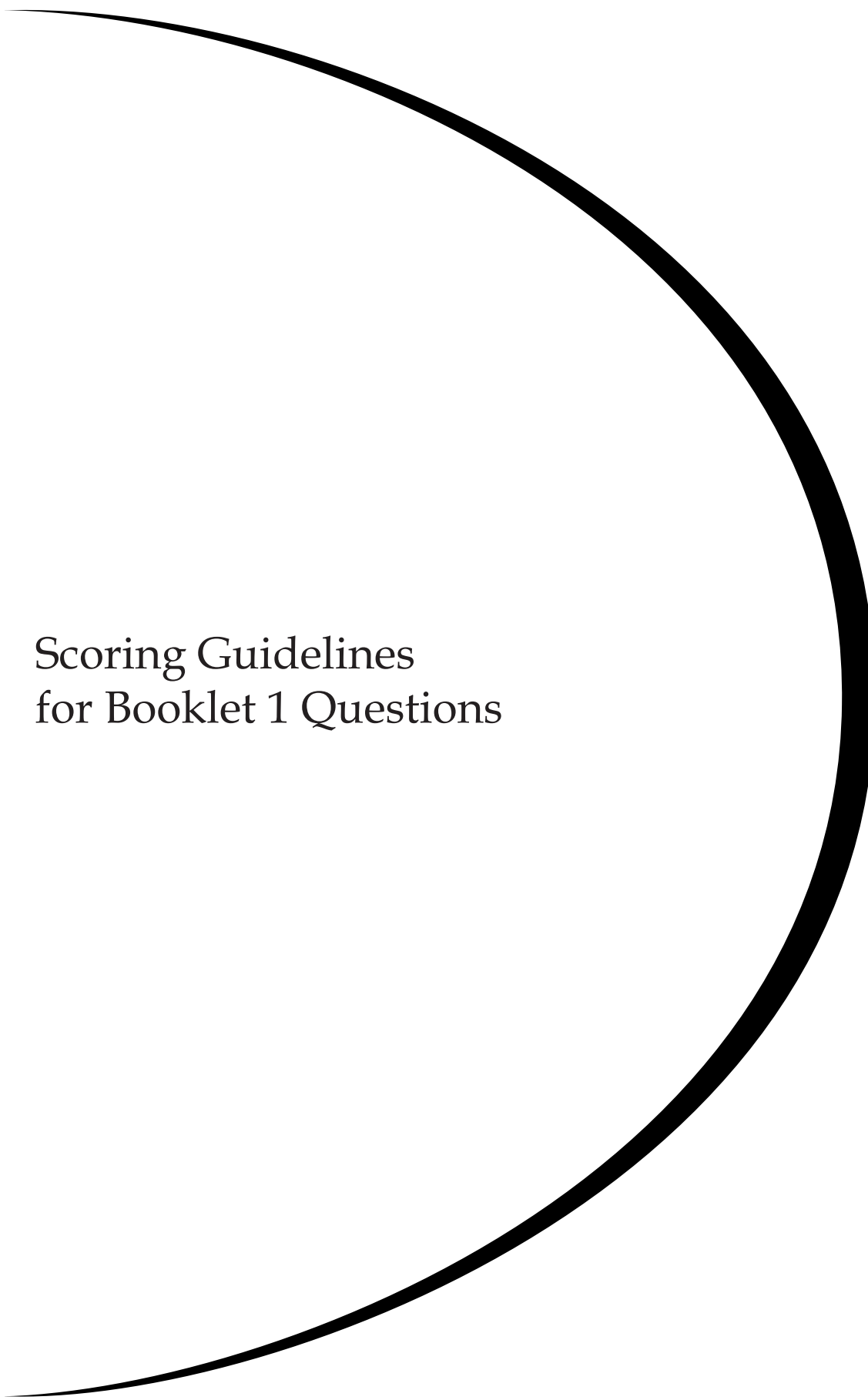
Example: A student has a preliminary mark of 72. The student committed two E1 errors ( $\frac{1}{2}$  mark deduction), four E7 errors ( $\frac{1}{2}$  mark deduction), and one E8 error ( $\frac{1}{2}$  mark deduction). Although seven communication errors were committed in total, there is a deduction of only  $1\frac{1}{2}$  marks.

| COMMUNICATION ERRORS / ERREURS DE COMMUNICATION                                                                                                                                                                                      |                                  |    |                                  |    |                                  |    |                       |     |                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----|----------------------------------|----|----------------------------------|----|-----------------------|-----|-----------------------|
| Shade in the circles below for a maximum total deduction of 5 marks ( $\frac{1}{2}$ mark deduction per error).<br>Noircir les cercles ci-dessous pour une déduction maximale totale de 5 points (déduction de 0,5 point par erreur). |                                  |    |                                  |    |                                  |    |                       |     |                       |
| E1                                                                                                                                                                                                                                   | <input checked="" type="radio"/> | E2 | <input type="radio"/>            | E3 | <input type="radio"/>            | E4 | <input type="radio"/> | E5  | <input type="radio"/> |
| E6                                                                                                                                                                                                                                   | <input type="radio"/>            | E7 | <input checked="" type="radio"/> | E8 | <input checked="" type="radio"/> | E9 | <input type="radio"/> | E10 | <input type="radio"/> |

Example: Marks assigned to the student

| Marks Awarded | Booklet 1 | Selected Response | Booklet 2 | Communication Errors (Deduct) | Total           |
|---------------|-----------|-------------------|-----------|-------------------------------|-----------------|
|               | 25        | 7                 | 40        | $1\frac{1}{2}$                | $70\frac{1}{2}$ |
| Total Marks   | 36        | 9                 | 45        | Maximum deduction of 5 marks  | 90              |





## Scoring Guidelines for Booklet 1 Questions

## Question 1

P1

Manitoba postal codes consist of three letters and three digits. Determine the total number of possible postal codes in Manitoba if the following characteristics must be met:

- The postal code must begin with the letter R.
- Only 18 letters of the alphabet can be used.
- The letters and digits must alternate.
- The letters and digits can be repeated.

### Solution

$$\frac{1 \cdot 10 \cdot 18 \cdot 10 \cdot 18 \cdot 10}{R} = 324\,000$$

1 mark

### Note:

The restriction of the first letter does not need to be shown.

### Exemplar 1

---

$$\underline{1} \cdot \underline{9} \cdot \underline{18} \cdot \underline{9} \cdot \underline{18} \cdot \underline{9}$$

$$= \underline{\underline{236196 \text{ ways}}}$$

---

½ out of 1

award full marks

– ½ mark for procedural error (9 digits instead of 10)

### Exemplar 2

---

$$\underline{1} \underline{17} \underline{15} \quad \underline{10} \underline{9} \underline{8} = 195840 \text{ ways}$$

---

½ out of 1

award full marks

– ½ mark for procedural error (non-repeating digits and letters)

### Exemplar 3

---

$$\underline{1!} \cdot \underline{10!} \cdot \underline{18!} \underline{10!} \underline{18!} \underline{10!}$$

---

0 out of 1

award full marks

– 1 mark for concept error (applying factorials)

## Question 2

R10

The percent of people who click on a link, in response to seeing an advertisement on YouTube, can be modeled by the formula:

$$R(t) = -(0.8)^{0.2t} + 0.6$$

where  $R(t)$  is the percentage, in decimal form, of people who click on the link

$t$  is the time in days.

Determine, algebraically, the number of days it will take for 45% of people who see the advertisement to click on the link.

### Solution

$$0.45 = -(0.8)^{0.2t} + 0.6$$

$$-0.15 = -(0.8)^{0.2t}$$

$$0.15 = (0.8)^{0.2t}$$

$$\log(0.15) = \log(0.8)^{0.2t}$$

$$\log(0.15) = 0.2t \cdot \log(0.8)$$

$$\frac{\log(0.15)}{0.2 \cdot \log(0.8)} = t$$

$$t = 42.508958\dots$$

$$t = 42.509 \text{ days}$$

½ mark for applying logarithms

1 mark for power law

½ mark for evaluating quotient of logarithms

**2 marks**

### Exemplar 1

---

$$\begin{aligned}45 &= -(0.8)^{0.2t} + 0.6 \\44.4 &= -(0.8)^{0.2t} \\ \log 44.4 &= -\log(0.8)^{0.2t} \\ \frac{\log 44.4}{-\log(0.8)} &= \frac{-0.2t \log(0.8)}{-\log(0.8)} \\ \frac{0.2t}{0.2} &= \frac{16.999}{0.2} \\ t &= 8.4 \text{ days}\end{aligned}$$

---

½ out of 2

- + 1 mark for power law
- + ½ mark for evaluating quotient of logarithms
- ½ mark procedural error in line 1 (substituting percentage as a whole number)
- ½ arithmetic error in line 6

### Exemplar 2

---

$$\begin{aligned}0.45 &= -(0.8)^{0.2t} + 0.6 \\ \log 0.45 &= -0.2t \log 0.8 + 0.6 \\ t &= 5.964 \text{ days}\end{aligned}$$

---

1 out of 2

- + 1 mark for power law

### Exemplar 3

---

$$\begin{aligned}(0.8)^{0.2t} &= 0.45 - 0.6 \\ 0.2t \log(0.8) &= \log(-0.15) \\ t &= \frac{\log(-0.15)}{0.2 \log(0.8)}\end{aligned}$$

---

1 out of 2

- + ½ mark for applying logarithms
- + 1 mark for power law
- ½ mark for procedural error line 1

### Question 3

P4

Determine and simplify the middle term in the binomial expansion of  $\left(\frac{5}{x} + 4x^3\right)^6$ .

#### Solution

$$t_4 = {}_6C_3 \left(\frac{5}{x}\right)^3 (4x^3)^3$$

2 marks (1 mark for  ${}_6C_3$ ;  $\frac{1}{2}$  mark for each consistent factor)

$$= 20 \left(\frac{125}{x^3}\right) (64x^9)$$

$$= 160\,000x^6$$

1 mark for simplification ( $\frac{1}{2}$  mark for coefficient;  $\frac{1}{2}$  mark for exponent)

**3 marks**

### Exemplar 1

---

$$t_{3+1} = {}_6C_3 \left(\frac{5}{x}\right)^{6-3} (4x^3)^3$$

$$\frac{6!}{(6-3)!3!} = 20$$

$$t_4 = 20 \left(\frac{5}{x}\right)^3 (4x^3)$$

$$t_4 = \frac{20}{1} \left(\frac{125}{x^3}\right) \left(\frac{262144 x^6}{1}\right)$$

$$= \boxed{655360000 x^3}$$

---

2½ out of 3

+ 1 mark for  ${}_6C_3$

+ 1 mark for consistent factors

+ ½ mark for exponent

### Exemplar 2

---

$$t = 4$$

$$k = 3$$

$$n = 7$$

$$a = \frac{5}{x}$$

$$b = 4x^3$$

$$t_{3+1} = {}_nC_3 \left(\frac{5}{x}\right)^{n-3} (4x^3)^3$$

$$t_4 = {}_nC_3 \left(\frac{5}{x}\right)^{n-3} (64x^9)$$

$$t_4 = 7C_3 \left(\frac{5}{x}\right)^{7-3} (64x^9)$$

$$t_4 = 35 \left(\frac{5}{x}\right)^4 (64x^9)$$

$$t_4 = 35 \left(\frac{625}{x^4}\right) (64x^9)$$

$n = \text{number of terms}$

$n$

$$t_4 = \frac{1400000x^9}{x^4}$$

$$t_4 = \frac{1400000}{x^{-5}}$$

---

2 out of 3

+ 1 mark for consistent factors

+ 1 mark for simplification

## Question 4

T5

Solve  $3\tan^2 x + 5\tan x - 6 = 0$ , over the interval  $[0, 2\pi]$ .

### Solution

$$\tan x = \frac{-5 \pm \sqrt{5^2 - 4(3)(-6)}}{2(3)}$$

$$\tan x = \frac{-5 \pm \sqrt{97}}{6}$$

1 mark for solving for  $\tan x$

$$\tan x = 0.808\,142\dots$$

$$\tan x = -2.474\,809\dots$$

$$x_r = 0.679\,686\dots$$

$$x_r = 1.186\,784\dots$$

$$x = 0.680, 3.821$$

$$x = 1.955, 5.096$$

2 marks for solving for  $x$  ( $\frac{1}{2}$  mark for each value)

**3 marks**

## Exemplar 1

$$\begin{aligned} 1 + \tan x &= x \\ 3 \tan^2 x + 5 \tan x - 6 &= 0 \\ 3x^2 + 5x - 6 &= 0 \end{aligned}$$
$$D = b^2 - 4ac$$
$$25 - 4(3)(-6)$$
$$D = 97$$
$$x = \frac{-b \pm \sqrt{D}}{2a}$$
$$x = \frac{-5 \pm \sqrt{97}}{2(3)}$$
$$x = \frac{-5 \pm \sqrt{97}}{6}$$
$$x = \frac{-5 - \sqrt{97}}{6}$$
$$x = -2.47$$
$$x = \frac{-5 + \sqrt{97}}{6}$$
$$x = 0.81$$
$$\tan x = -2.47$$
$$\tan x = 0.81$$
$$x = 0.68 \text{ rad}, 3.822 \text{ rad}, 1.186 \text{ rad}, 4.327 \text{ rad}$$

2 out of 3

+ 1 mark for solving for  $\tan x$

+ 1 mark for solving for  $x$

E7 (notation error in line 1)

E6 (rounding too early)

## Exemplar 2

$$\tan x = \frac{-5 \pm \sqrt{5^2 - 4(3)(-6)}}{2(3)}$$

$$\tan x = \frac{-5 \pm \sqrt{25 + 72}}{6}$$

$$\tan x = \frac{-5 \pm \sqrt{97}}{6}$$

$$\left\{ 0.808, -2.475 \right\}$$

1 out of 3

+ 1 mark for solving for  $\tan x$

### Exemplar 3

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-5 \pm \sqrt{(5)^2 - 4(3)(-6)}}{2(3)}$$

$$x = \frac{-5 \pm \sqrt{25 + 72}}{6}$$

$$x = \frac{-5 \pm \sqrt{97}}{6}$$

|                 |                   |             |                   |
|-----------------|-------------------|-------------|-------------------|
| $x = -2.475$    |                   | $x = 0.808$ |                   |
| Q II            | Q III             | Q I         | Q III             |
| $\pi - 2.475$   | $x = \pi - 2.475$ | $x = 0.808$ | $x = \pi + 0.808$ |
| $\approx 5.616$ | $\approx 3.808$   |             | $\approx 3.950$   |

$$\{x = 0.808, 3.808, 3.950, 5.616\}$$

#### 1 out of 3

- + 1 mark for solving for  $\tan x$
- + 1 mark for solving for  $x$  (consistent with their reference angle in right branch)
- 1 mark for concept error (using  $\tan x$  as a reference angle)

### Exemplar 4

$$3\tan^2 x + 5\tan x - 6 = 0$$

$$x = \frac{(-5) \pm \sqrt{5^2 - 4(3)(-6)}}{2(3)}$$

$$\frac{-5 \pm \sqrt{97}}{6}$$

$$\frac{-5 + \sqrt{97}}{6} = 0.608142861$$

$$= 38.94316675$$

$$\boxed{= 38.943^\circ}$$

$$\frac{-5 - \sqrt{97}}{6} = -2.474809634$$

$$= -67.99777085$$

$$= -67.998^\circ$$

$$-67.998 + 360^\circ$$

$$= 292.0022292$$

$$\boxed{= 292.002^\circ}$$

$$\{x = 38.943^\circ, 292.002^\circ\}$$

#### 2 out of 3

- + 1 mark for solving for  $\tan x$
- + 1 mark for solving for  $x$  (consistent with their answer)
- E2 (changing an equation into an expression)
- E5 (answer stated in degrees instead of radians)
- E7 (notation errors)

### Exemplar 5

---

$$(3\tan x + 2)(\tan x - 3) = 0$$

$$\begin{array}{cc} \text{Q2,4} & \text{Q1 and 3} \\ \tan x = -\frac{2}{3} & \tan x = 3 \end{array}$$

$$x_R = 0.588 \quad x_R = 1.249$$

$$\begin{array}{l} x = \pi - 0.588 \\ \boxed{x = 2.554} \end{array}$$

$$\boxed{x = 1.249}$$

$$\begin{array}{l} x = 2\pi - 0.588 \\ \boxed{x = 5.695} \end{array}$$

$$\begin{array}{l} x = \pi + 1.249 \\ \boxed{x = 4.391} \end{array}$$

---

2 out of 3

+ 2 marks for solving for x

## Question 5

T1

Given  $\theta = -80^\circ$ ,

a) determine the measure of  $\theta$ , in radians.

b) state all the coterminal angles of  $\theta$ .

### Solution

a)  $\theta = -80^\circ \left( \frac{\pi}{180^\circ} \right)$

1 mark

$$\theta = -\frac{4\pi}{9}$$

or

$$\theta = -1.396$$

b)  $\theta = -80^\circ + 360^\circ k, k \in \mathbb{Z}$

or

$$\theta = -\frac{4\pi}{9} + 2k\pi, k \in \mathbb{Z}$$

or

$$\theta = -1.396 + 2k\pi, k \in \mathbb{Z}$$

1 mark

### Exemplar 1

---

a)  $\theta = -80^\circ$

$$\theta = -80^\circ \cdot \frac{\pi}{180^\circ} = \frac{-80\pi}{180^\circ} = \boxed{-\frac{4\pi}{9} \text{ rad}}$$

1 out of 1

b)  $-\frac{22\pi}{9}, \frac{14\pi}{9}, \frac{32\pi}{9}$

0 out of 1

### Exemplar 2

---

a)  $-80 \cdot \frac{\pi}{180} = \boxed{-1.396}$

1 out of 1

b)  $-1.396 + 2\pi k$

½ out of 1

award full marks

– ½ mark for procedural error

### Exemplar 3

---

a)  $-\frac{80^\circ}{180} \cdot \frac{\pi}{2.25} = \boxed{-\frac{\pi}{2.25}}$

1 out of 1

award full marks

E1 (final answer given as a complex fraction)

b)  $-\frac{\pi}{2.25} + 2\pi k, k \in \mathbb{R}$

½ out of 1

award full marks

– ½ mark for procedural error

## Question 6

R12

State a possible value of  $n$  if the polynomial function  $p(x) = (x+1)^n(x-4)^2$  has a range of  $(-\infty, \infty)$ .

### Solution

$$n = 1$$

or

$$n = 3$$

1 mark

### Note:

Accept any odd whole number for  $n$ .

### Exemplar 1

---

No even numbers

3, 9 or others

---

1 out of 1

### Exemplar 2

---

$$(x-4)^2 = x^2 - 8x + 16$$

this gives  $p(x)$  a range of  $[0, +\infty)$

in order for it to have a range of  $(-\infty, \infty)$   
we need to have  $x$  to an odd degree.

$$(x+1)^{n=1} (x^2 - 8x + 16)$$

$$x^3 - 8x^2 + 16x + x^2 - 8x + 16$$

$$x^3 - 7x^2 + 8x + 16$$

this gives us a polynomial function where the range is  
 $(-\infty, +\infty)$ , therefore a possible value for  $n=1$ .

---

1 out of 1

## Question 7

R2, R3, R5

Describe the transformations used to obtain the graph of the function  $y = -2f(x + 8) - 5$  from the graph of  $y = f(x)$ .

### Solution

The graph of  $y = f(x)$  is reflected over the  $x$ -axis, vertically stretched by a factor of 2, translated left 8 units and down 5 units.

- 1 mark for vertical reflection
- 1 mark for vertical stretch
- 1 mark for horizontal translation
- 1 mark for vertical translation

**4 marks**

### Note:

Deduct a maximum of 1 mark for concept error of incorrect order of vertical transformations.

### Exemplar 1

---

8 left

5 down

vertical stretch  $\times 2$   
vertical change '4'

double  $\downarrow$

---

2 out of 4

+ 1 mark for horizontal translation

+ 1 mark for vertical translation

### Exemplar 2

---

The graph had a;

vertical stretch of  $-2$ ,  
horizontal shift 8 spaces to the left  
vertical shift 5 spaces down

---

3 out of 4

+ 1 mark for vertical stretch

+ 1 mark for horizontal translation

+ 1 mark for vertical translation

### Exemplar 3

---

$$y = a(b(x \pm h)) + k$$

$$(x, y) \mapsto (x + 8, -2y - 5)$$

horizontal translation 8 units positive/right

vertical reflection by factor of  $-2$  over the  $x$  axis  
and vertical translation of 5 units down/

---

2 out of 4

+ 1 mark for vertical reflection

+ 1 mark for vertical translation

## Question 8

R14

State the domain and range of the function  $f(x) = \frac{-2}{x^2}$ .

### Solution

Domain:  $\{x|x \neq 0, x \in \mathbb{R}\}$  or  $(-\infty, 0) \cup (0, \infty)$

1 mark for domain

Range:  $\{y|y < 0, y \in \mathbb{R}\}$  or  $(-\infty, 0)$

1 mark for range

**2 marks**

### Exemplar 1

---

Domain:  $\{x \in \mathbb{R} \mid x \neq 0\}$

Range:  $\{y \in \mathbb{R} \mid y \neq -2\}$

---

**1 out of 2**

+ 1 mark for domain

### Exemplar 2

---

Domain:  $\{x \in \mathbb{R}, x \neq -2\}$

Range:  $\{y \in \mathbb{R}, y \neq 0\}$

---

**0 out of 2**

### Exemplar 3

---

Domain:  $(-\infty, 0) \cup (0, \infty)$

Range:  $(-\infty, 0) \cup (0, \infty)$

---

**1 out of 2**

+ 1 mark for domain

## Question 9

P4

Given the following row of Pascal's Triangle, state the values in the next row.

1   5   10   10   5   1

**Solution**

1   6   15   20   15   6   1

1 mark

### Exemplar 1

---

1      6      15      15      6      1

---

0 out of 1

### Exemplar 2

---

6      15      20      15      6

---

0 out of 1

## Question 10

R6

Given  $f(x) = 3 - 2x$ , determine the equation of  $f^{-1}(x)$ .

### Solution

$$\begin{aligned}\text{Let } y &= f(x) \\ y &= 3 - 2x\end{aligned}$$

To determine the inverse of  $f(x)$ , switch  $x$  and  $y$ .

$$\begin{aligned}x &= 3 - 2y && \text{1 mark for switching } x \text{ and } y \\ 2y &= 3 - x \\ y &= \frac{3 - x}{2} && \frac{1}{2} \text{ mark for solving for } y \\ f^{-1}(x) &= \frac{3 - x}{2} && \frac{1}{2} \text{ mark for writing equation in terms of } f^{-1}(x)\end{aligned}$$

**2 marks**

### Exemplar 1

---

$$f(x) = 3 - 2x$$

$$y = 3 - 2x$$

$$-\frac{x}{-2} = \frac{3-2y}{-2}$$

$$\frac{-3x}{-2} = \frac{-2y}{-2}$$

$$\frac{3}{2}x = y$$

$$f^{-1}(x) = \frac{3}{2}x$$

---

1½ out of 2

+ 1 mark for switching x and y

+ ½ mark for writing the equation in terms of  $f^{-1}(x)$

### Exemplar 2

---

$$f(x) = y$$

$$y = 3 - 2x$$

$$x = 3 - 2y$$

$$\frac{x+2}{-3} = y$$

---

1 out of 2

+ 1 mark for switching x and y

### Exemplar 3

---

$$y = 3 - 2x$$

$$x = \frac{3-y}{2}$$

$$\frac{x-3}{-2} = \frac{-y}{-2}$$

$$\frac{x-3}{-2} = f^{-1}(x)$$

---

2 out of 2

award full marks

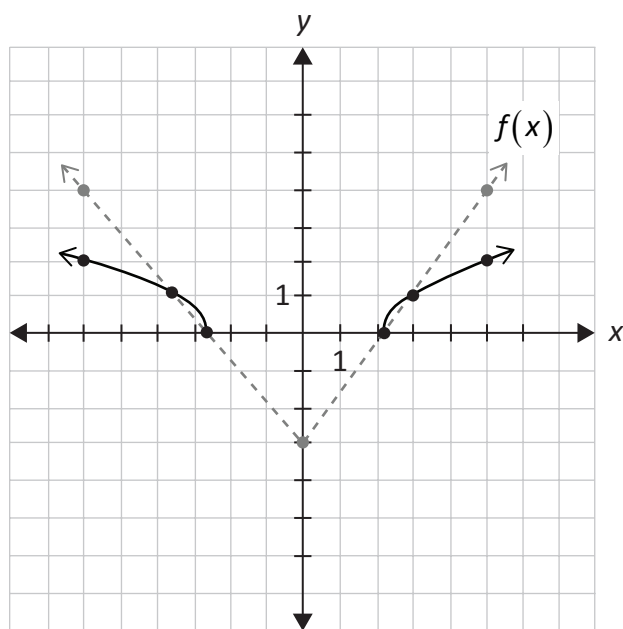
E7 (notation error in line 2)

## Question 11

R13

Given the graph of  $y = f(x)$ , sketch the graph of  $y = \sqrt{f(x)}$ .

### Solution



1 mark for restricting domain

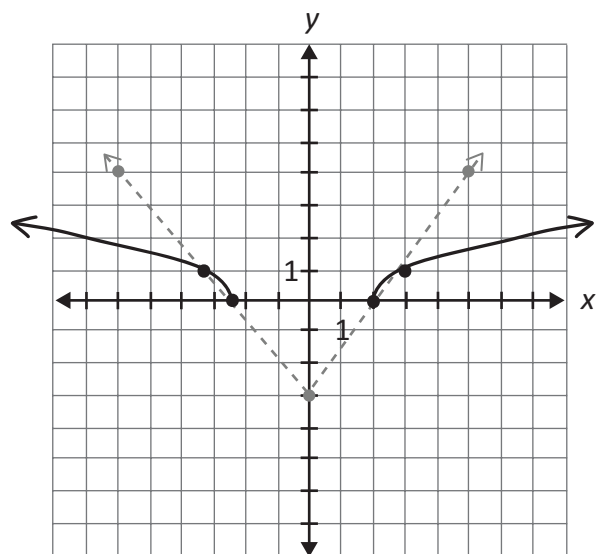
$\frac{1}{2}$  mark for shape between invariant points,  $\{0 \leq y \leq 1\}$

$\frac{1}{2}$  mark for shape above invariant points,  $\{y \geq 1\}$

**2 marks**

## Exemplar 1

---



---

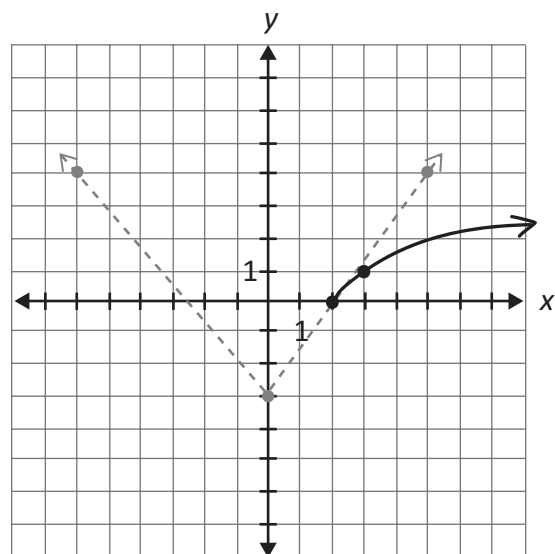
**1½ out of 2**

+ 1 mark for restricting domain

+ ½ mark for shape between invariant points

## Exemplar 2

---



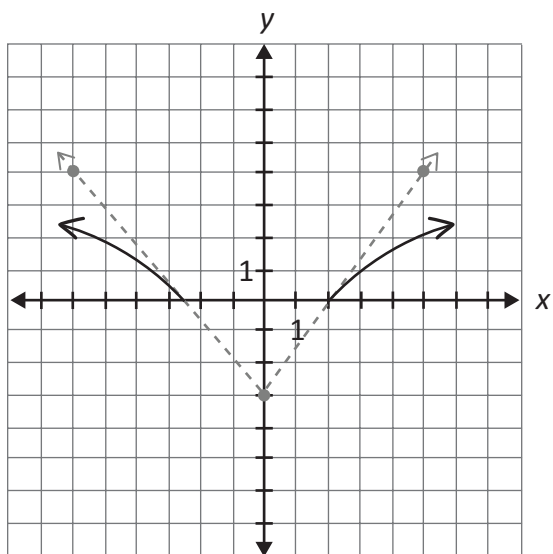
---

**½ out of 2**

+ ½ mark for shape above invariant points

### Exemplar 3

---



---

**1 out of 2**

+ 1 mark for restricting domain

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## Question 12

T6

Prove the identity for all permissible values of  $x$ .

$$\frac{\csc^2 x + \sec^2 x}{\tan x + \cot x} = \csc x \sec x$$

### Solution

#### Method 1

| Left-Hand Side                                                                                    | Right-Hand Side |
|---------------------------------------------------------------------------------------------------|-----------------|
| $\frac{\frac{1}{\sin^2 x} + \frac{1}{\cos^2 x}}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}}$   | $\csc x \sec x$ |
| $\frac{\frac{\cos^2 x + \sin^2 x}{\sin^2 x \cos^2 x}}{\frac{\cos^2 x + \sin^2 x}{\cos x \sin x}}$ |                 |
| $\frac{\cancel{\cos x} \cancel{\sin x}}{\sin^{\cancel{2}} x \cos^{\cancel{2}} x}$                 |                 |
| $\frac{1}{\sin x \cos x}$                                                                         |                 |
| $\csc x \sec x$                                                                                   |                 |

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

**3 marks**

## Question 12

T6

### Method 2

| Left-Hand Side                                                                                                                                                                                                                                                                                                                                                                                         | Right-Hand Side |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| $\frac{1}{\sin^2 x} + \frac{1}{\cos^2 x}$ $\frac{\tan x}{1} + \frac{1}{\tan x}$ $\frac{\cos^2 x + \sin^2 x}{\sin^2 x \cos^2 x}$ $\frac{\tan^2 x + 1}{\tan x}$ $\frac{1}{\sin^2 x \cos^2 x}$ $\frac{\sec^2 x}{\tan x}$ $\frac{\tan x}{\sin^2 x \cancel{\cos^2 x} \cancel{\sec^2 x}}$ $\frac{\cancel{\sin x}}{\cos x}$ $\frac{\cos x}{\sin^2 x}$ $\frac{1}{1}$ $\frac{1}{\sin x \cos x}$ $\csc x \sec x$ | $\csc x \sec x$ |

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

**3 marks**

## Exemplar 1

| Left-Hand Side                                                                                                                                                                                    | Right-Hand Side |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| $\frac{\csc^2 x + \sec^2 x}{\tan x + \cot x}$                                                                                                                                                     |                 |
| $\frac{1 + \cot^2 x + \tan^2 x + 1}{\frac{\sin x}{\cos x} + \frac{1}{\sin x}}$                                                                                                                    |                 |
| $\frac{(1 + \cot^2 \theta) + (\tan^2 \theta + 1)}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}}$                                                                                                 |                 |
| $\frac{\frac{1}{\tan^2 x} + \frac{\sin^2 x}{\cos^2 x} + 2}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}}$                                                                                        |                 |
| $\frac{\frac{\cos^2 x}{\sin^2 x} + \frac{\sin^2 x}{\cos^2 x} + 2}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}}$                                                                                 |                 |
| $\frac{\frac{\cos^2 x}{\sin^2 x} + \frac{\sin^2 x}{\cos^2 x} + 2}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}} = \frac{\frac{\cos x}{\sin x} + \frac{\sin x}{\cos x} + 2}{\cot x + \tan x + 2}$ |                 |

1 out of 3

+ 1 mark for correct substitution of identities

E3 (variable being introduced without being defined in line 3)

## Exemplar 2

| Left-Hand Side                                                                                                              | Right-Hand Side                                 |
|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| $\frac{\csc^2 x + \sec^2 x}{\tan x + \cot x}$                                                                               | $\csc x \sec x$                                 |
| $\frac{\frac{1}{\sin^2 x} + \frac{1}{\cos^2 x}}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}}$                             | $\frac{1}{\sin x} \cdot \frac{1}{\cos x}$       |
| $\frac{\frac{1}{\sin^2 x} + \frac{1}{\cos^2 x}}{\frac{(\sin x)(\sin x)(\cos x)(\cos x)}{(\sin x)(\cos x)(\sin x)(\cos x)}}$ | $\frac{1}{\sin x \cos x}$                       |
| $\frac{\frac{1}{\sin^2 x} + \frac{1}{\cos^2 x}}{\frac{1}{\sin x \cos x}}$                                                   |                                                 |
| $\frac{1}{\sin^2 x \cos^2 x}$                                                                                               |                                                 |
| $\frac{1}{\sin^2 x \cos^2 x + \sin^2 x \cos^2 x}$                                                                           |                                                 |
| <p>Top:</p> $\frac{1(\cos^2 x)}{(\sin^2 x)(\cos^2 x)} + \frac{1(\sin^2 x)}{(\sin^2 x)(\cos^2 x)}$                           | $\frac{\sin^2 x + \cos^2 x}{\sin^2 x \cos^2 x}$ |
| $\frac{\sin^2 x + \cos^2 x}{(\sin^2 x)(\cos^2 x)}$                                                                          |                                                 |
| $\frac{1}{\sin^2 x \cos^2 x}$                                                                                               |                                                 |

1 out of 3

+ 1 mark for correct substitution of identities

E2 (equating the two sides when proving an identity)

### Question 13

R9

Justify that the graph of  $f(x) = e^{x+1} + 5$  does not have an x-intercept.

#### Solution

$$0 = e^{x+1} + 5$$

$$-5 = e^{x+1}$$

$e^{x+1} > 0$ , therefore there is no solution.

or

The range of the graph is  $\{y \mid y > 5, y \in \mathbb{R}\}$ .

or

The horizontal asymptote is  $y = 5$  and it is an increasing exponential function above the asymptote.

1 mark

### Exemplar 1

---

because if  $f(x) = e^{x+1}$  graph has horizontal asymptote  
on the  $x$ -axis, which means you can't touch the  $x$ -axis.

---

½ out of 1

award full marks

– ½ mark for lack of clarity in the explanation

### Exemplar 2

---

the graph starts at  $(0, 5)$  and trends upwards

---

0 out of 1

### Exemplar 3

---

because it's an exponential graph  
and the asymptote on  $y=5$ , doesn't  
let the line pass through the  $x$ -axis

---

½ out of 1

award full marks

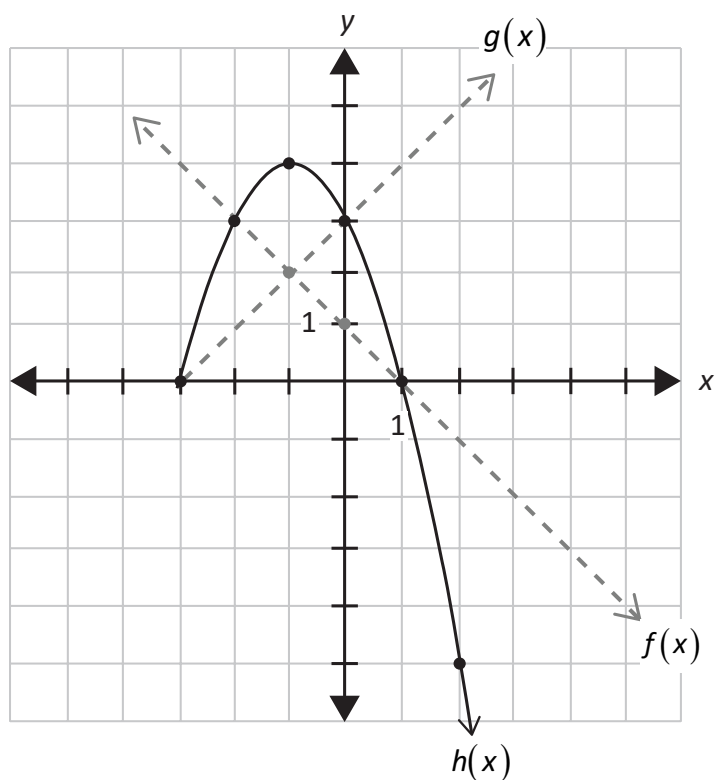
– ½ mark for lack of clarity in the explanation

## Question 14

R1

Given the graphs of  $f(x)$  and  $g(x)$ , sketch the graph of  $h(x) = f(x) \cdot g(x)$ .

### Solution

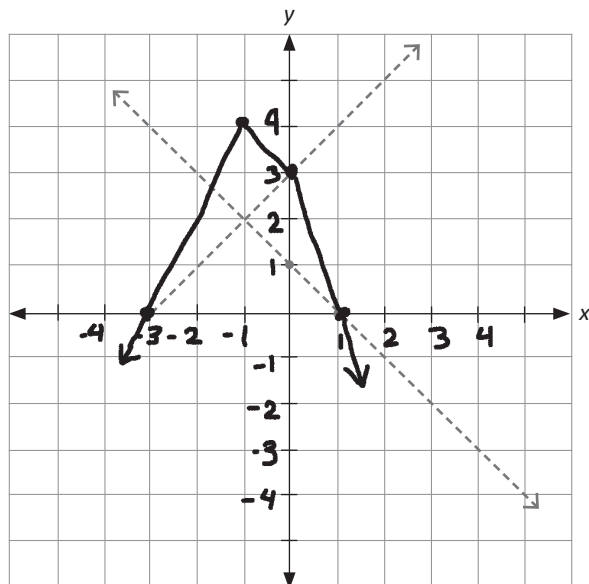


1 mark for operation of multiplication  
1 mark for restricted domain

**2 marks**

## Exemplar 1

---



---

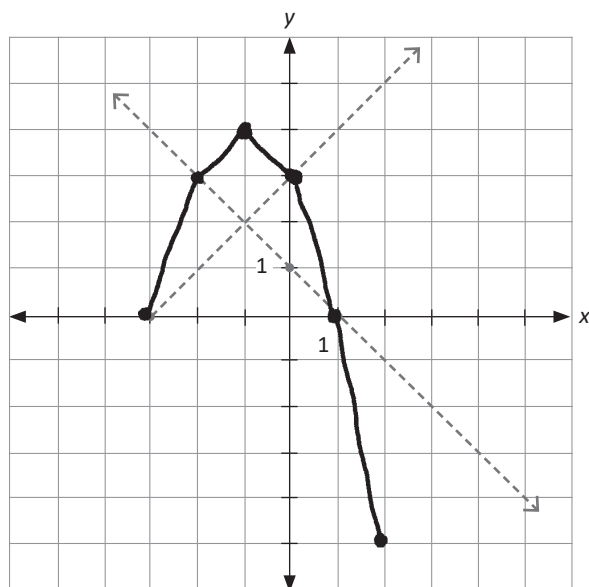
$\frac{1}{2}$  out of 2

+ 1 mark for operation of multiplication

–  $\frac{1}{2}$  mark for incorrect shape of the graph

## Exemplar 2

---



---

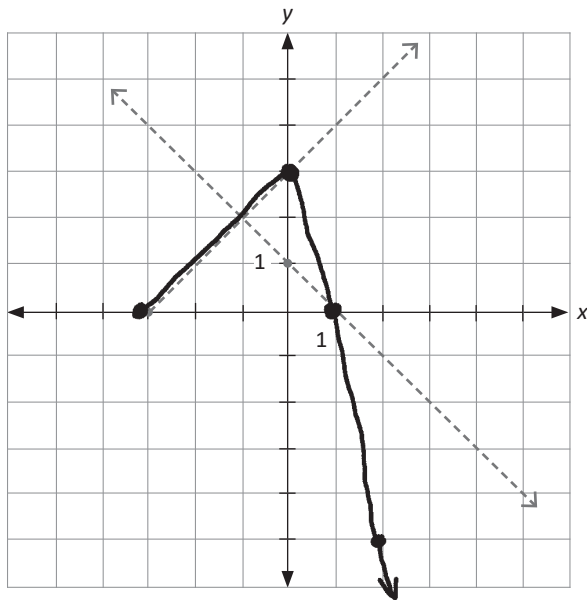
2 out of 2

award full marks

E9 (arrowhead omitted on RHS of graph)

### Exemplar 3

---



---

**1 out of 2**

+ 1 mark for restricted domain

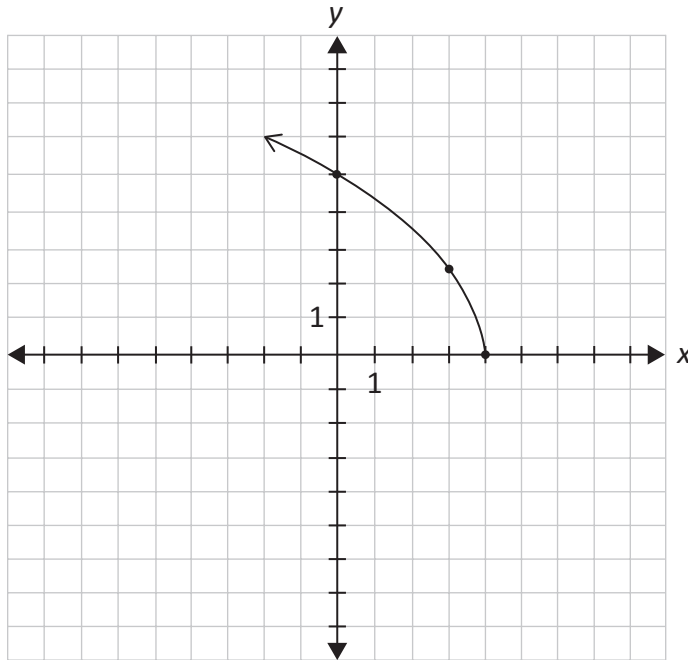
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### Question 15

R13

Sketch the graph of  $f(x) = \frac{5}{2}\sqrt{-(x-4)}$ .

#### Solution

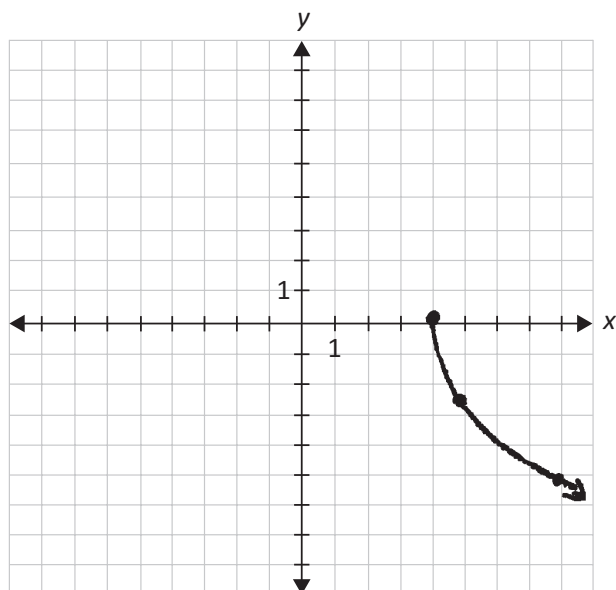


- 1 mark for shape of a radical function
- 1 mark for vertical stretch
- 1 mark for horizontal reflection
- 1 mark for horizontal translation

**4 marks**

## Exemplar 1

---



---

**3 out of 4**

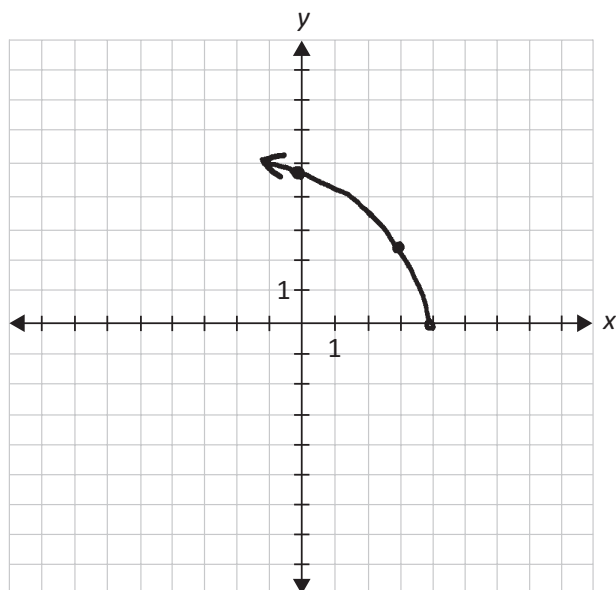
+ 1 mark for shape of a radical function

+ 1 mark for vertical stretch

+ 1 mark for horizontal translation

## Exemplar 2

---



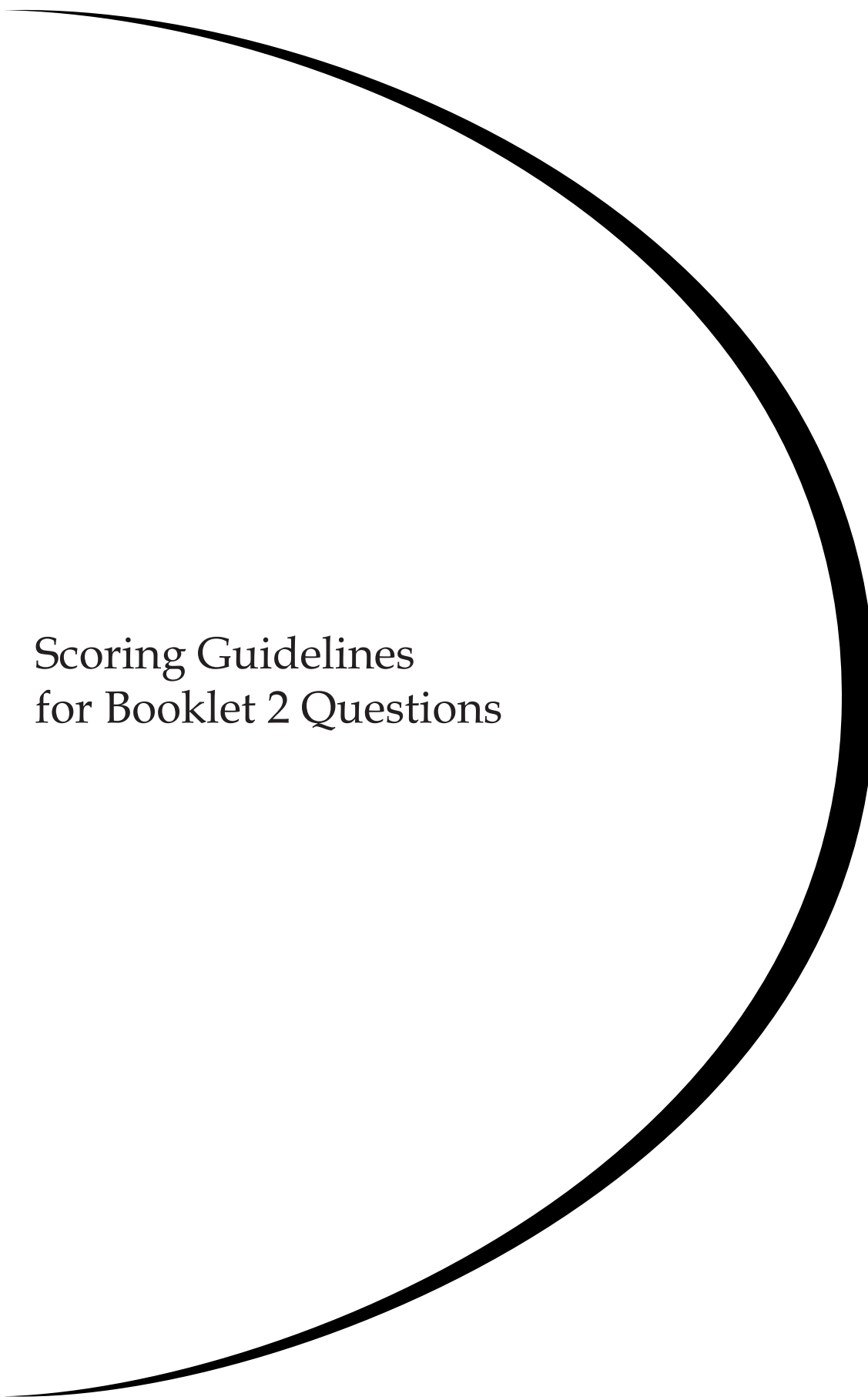
---

**3½ out of 4**

award full marks

– ½ mark for procedural error (one incorrect point)

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## Scoring Guidelines for Booklet 2 Questions

# Answer Key for Selected Reponse Questions

| Question | Answer | Learning Outcome |
|----------|--------|------------------|
| 16       | B      | R11              |
| 17       | D      | R5               |
| 18       | B      | R7               |
| 19       | C      | R14              |
| 20       | C      | P2               |
| 21       | B      | R11              |
| 22       | D      | T4               |
| 23       | C      | R8               |
| 24       | A      | T1               |

**Question 16****R11**

Identify the number of  $x$ -intercepts on the graph of the polynomial function,

$$p(x) = 3x(x - 8)(x^2 + 5).$$

a. 1

b. 2

c. 3

d. 4

**Question 17****R5**

Identify the equation that represents the graph of  $y = x^2 - 3$  after a reflection over the  $x$ -axis.

a.  $y = -x^2 - 3$

b.  $y = x^2 - 3$

c.  $y = x^2 + 3$

d.  $y = -x^2 + 3$

**Question 18****R7**

Given  $2^{\log_2 3} = x$ , identify the value of  $x$ .

a. 1

b. 3

c. 6

d. 8

### Question 19

R14

Identify the y-intercept on the graph of  $y = \frac{x-3}{x^2-5x+6}$ .

a.  $-3$

b.  $-2$

c.  $-\frac{1}{2}$

d.  $3$

### Question 20

P2

Identify the total number of arrangements for 10 students and 2 teachers to sit in a row if the teachers must sit together.

a.  $10!2!$

b.  $11!$

c.  $11!2!$

d.  $12!$

### Question 21

R11

Identify which expression represents the remainder when the polynomial  $P(x)$  is divided by  $(x-4)$ .

a.  $P(-4)$

b.  $P(4)$

c.  $P(x-4)$

d.  $P(x+4)$

## Question 22

T4

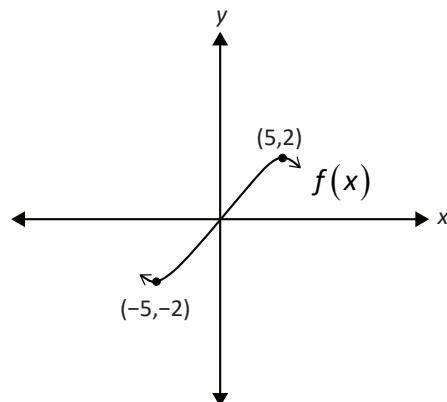
Identify the period of the graph of the sinusoidal function.

a.  $10\pi$

b. 10

c.  $20\pi$

d. 20



## Question 23

R8

Identify an equivalent expression for  $\log_3 9 + \log_3 5$ .

a.  $\log_3 14$

b.  $2\log_3 5$

c.  $\log_3 45$

d.  $\log_3 \left( \frac{9}{5} \right)$

## Question 24

T1

Identify the coterminal angle(s) of  $\theta = -\frac{11\pi}{10}$  over the interval  $[-3\pi, \pi]$ .

a.  $\theta = \frac{9\pi}{10}$

b.  $\theta = -\frac{31\pi}{10}, -\frac{21\pi}{10}$

c.  $\theta = -\frac{31\pi}{10}, \frac{9\pi}{10}$

d.  $\theta = -\frac{21\pi}{10}, \frac{9\pi}{10}$

## Question 25

R11

Express  $p(x) = x^3 - 13x - 12$  in completely factored form.

### Solution

$$p(-1) = (-1)^3 - 13(-1) - 12$$

$$p(-1) = 0$$

$\therefore (x + 1)$  is a factor

1 mark for identifying one possible zero of  $P(x)$

|    |  |   |    |     |     |
|----|--|---|----|-----|-----|
| -1 |  | 1 | 0  | -13 | -12 |
|    |  |   | -1 | 1   | 12  |
|    |  | 1 | -1 | -12 | 0   |

1 mark for synthetic division (or equivalent strategy)

$$p(x) = (x + 1)(x^2 - x - 12)$$

$$p(x) = \underline{(x + 1)(x - 4)(x + 3)}$$

1 mark for consistent product of factors

**3 marks**

### Exemplar 1

---

Test :  $(x+1)$

$$\begin{aligned}P(-1) &= (-1)^3 - 13(-1) - 12 \\&= -1 + 13 - 12 \\&= 0\end{aligned}$$

$$\begin{array}{r|rrrr} -1 & 1 & 0 & -13 & -12 \\ & & -1 & 1 & 12 \\ \hline & 1 & -1 & -12 & 0 \end{array}$$

$$\begin{aligned}P(x) &= (x^2 - x - 12) \\&= (x - 4)(x + 3)\end{aligned}$$

---

2½ out of 3

award full marks

– ½ mark for procedural error (not including the first factor in final answer)

### Exemplar 2

---

$$\begin{array}{r|rrrr} -3 & 1^3 & 0^2 & -13^1 & -12^0 \\ & \downarrow & & & \\ & -3 & 9 & 12 & \\ \hline & 1 & -3 & -4 & \boxed{0} \end{array}$$

$$x^2 - 3x - 4$$

$$\begin{aligned}ac &= -4 \Rightarrow -4 \cdot 1 \\ b &= -3 \Rightarrow -4 + 1\end{aligned}$$

$$P(x) = \underline{(x-3)(x-4)(x+1)}$$

---

2½ out of 3

award full marks

– ½ mark for procedural error (using the wrong sign when identifying the first factor)

## Question 26

T5

Verify, by substitution, that the equation  $\frac{\cos^2 \theta}{\sin^2 \theta} = \frac{2 \csc \theta}{\sec^2 \theta}$  is true for  $\theta = \frac{\pi}{6}$ .

### Solution

| Left-Hand Side                                                              | Right-Hand Side                                                             |                                                               |
|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------|
| $\frac{\cos^2\left(\frac{\pi}{6}\right)}{\sin^2\left(\frac{\pi}{6}\right)}$ | $\frac{2 \csc\left(\frac{\pi}{6}\right)}{\sec^2\left(\frac{\pi}{6}\right)}$ |                                                               |
| $\frac{\left(\frac{\sqrt{3}}{2}\right)^2}{\left(\frac{1}{2}\right)^2}$      | $\frac{2(2)}{\left(\frac{2}{\sqrt{3}}\right)^2}$                            | 2 marks for exact values (½ mark for each)                    |
| $\frac{\frac{3}{4}}{\frac{1}{4}}$                                           | $\frac{2(2)}{\left(\frac{4}{3}\right)}$                                     |                                                               |
|                                                                             | $\frac{12}{4}$                                                              |                                                               |
| 3                                                                           | 3                                                                           | 1 mark for simplification<br>(½ mark for LHS; ½ mark for RHS) |

**3 marks**

### Exemplar 1

| Left-Hand Side                                                                                       | Right-Hand Side                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{\cos^2 \theta}{\sin^2 \theta}$ $\frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}}$ $\frac{2\sqrt{3}}{2}$ | $\frac{2 \frac{1}{\sin \theta}}{\frac{1}{\cos^2 \theta}}$ $2 \frac{\cos^2 \theta}{\sin \theta}$ $\frac{2 \frac{\sqrt{3}}{2}}{\frac{1}{2}}$ $\frac{4\sqrt{3}}{2}$ |
| $LS \neq RS$                                                                                         |                                                                                                                                                                  |

2½ out of 3

award full marks

– ½ mark for procedural error (not squaring values)

## Exemplar 2

| Left-Hand Side                                           | Right-Hand Side                                      |
|----------------------------------------------------------|------------------------------------------------------|
| $\frac{\cos^2(\frac{\sqrt{3}}{2})}{\sin^2(\frac{1}{2})}$ | $\frac{2\csc(\frac{\pi}{4})}{\sec^2(\frac{\pi}{3})}$ |
| $\frac{(\frac{\sqrt{3}}{2})^2}{(\frac{1}{2})^2}$         | $\frac{2(\frac{2}{1})}{(\frac{2}{\sqrt{3}})^2}$      |
| $\frac{\frac{3}{4}}{\frac{1}{4}}$                        | $\frac{\frac{4}{1}}{\frac{4}{3}}$                    |
| $\frac{3}{4} \times \frac{4}{1} = \frac{12}{4}$          | $\frac{4}{1} \times \frac{3}{4} = \frac{12}{4}$      |
| $\frac{3}{1}$                                            | $\frac{3}{1}$                                        |
| 3                                                        | 3                                                    |

2½ out of 3

award full marks

– ½ mark for procedural error in line 1

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## Question 27

R9

Given  $f(x) = \log_3 x + 2$ ,

a) determine the x-intercept of  $f(x) = \log_3 x + 2$ .

b) sketch the graph of  $f(x) = \log_3 x + 2$ .

### Solution

a)

$$0 = \log_3 x + 2$$

$$-2 = \log_3 x$$

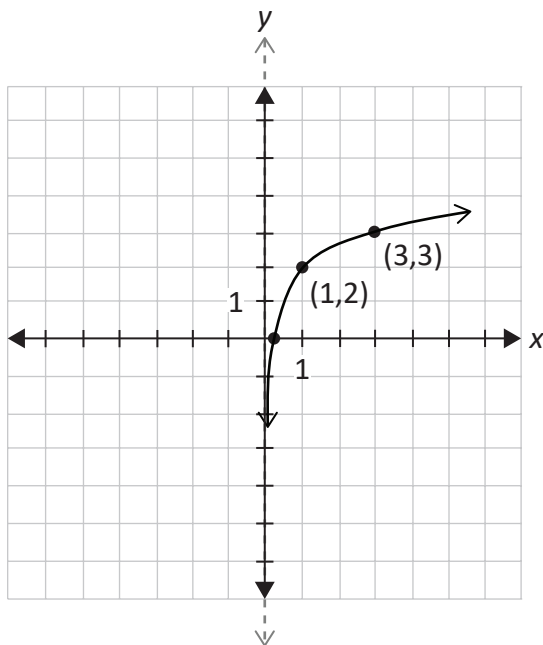
$$3^{-2} = x$$

$$\frac{1}{3^2} = x$$

$$\frac{1}{9} = x$$

1 mark

b)



1 mark for shape of a logarithmic function  
1 mark for vertical translation

2 marks

### Exemplar 1

---

a)

$$0 = \log_3 x + 2$$

$$-2 = \log_3 x$$

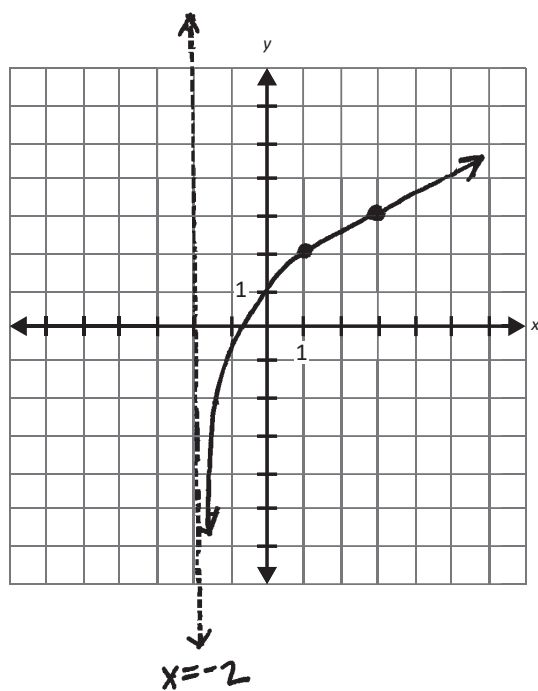
$$3^{-2} = x$$

$$x = \frac{1}{9}$$

---

1 out of 1

b)



---

1 out of 2

award full marks

– 1 mark for concept error (shifting the asymptote left 2)

## Exemplar 2

---

a)

$$0 = \log_3 x + 2$$

$$-2 = \log_3 x$$

$$3^{-2} = x$$

$$\frac{1}{9} = x$$

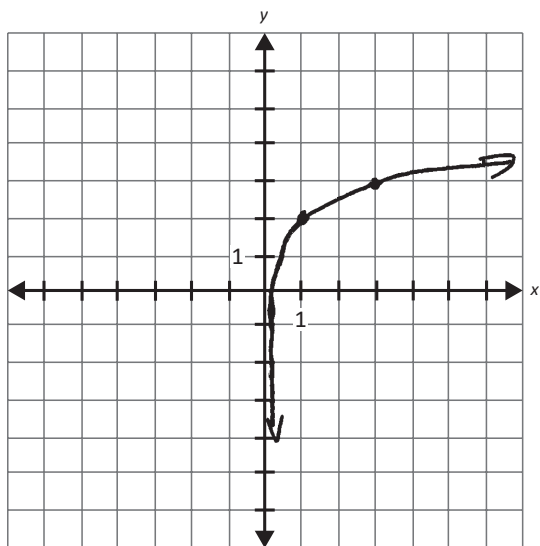
---

½ out of 1

award full marks

– ½ mark for arithmetic error

b)



---

2 out of 2

award full marks

E10 (asymptote omitted but still implied)

### Exemplar 3

---

a)

For x-int :  $y = 0$

$$\log_3 x + 2 = 0$$

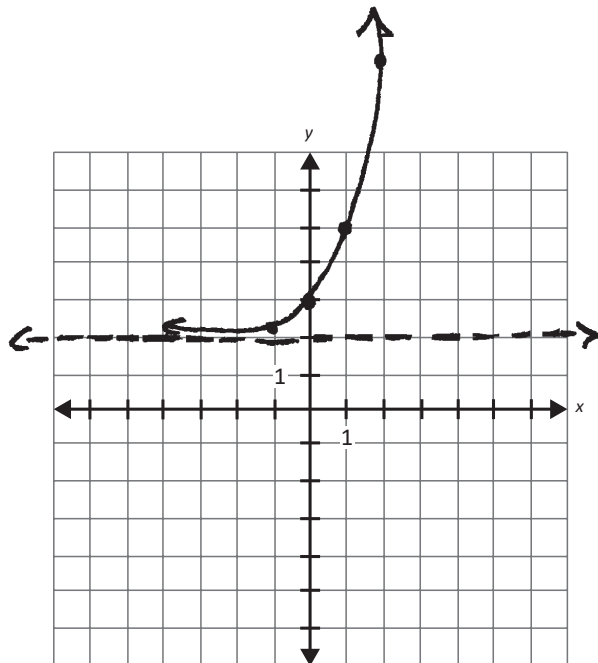
$$\log_3 x = -2$$

log can not be negative  
So, no x-intercept.

---

0 out of 1

b)



---

1 out of 2

+ 1 mark for vertical translation

## Question 28

T3

Evaluate.

$$\sec^2\left(\frac{5\pi}{4}\right) \cdot \tan^2\left(-\frac{2\pi}{3}\right)$$

### Solution

$$\left(-\frac{2}{\sqrt{2}}\right)^2 \left(\frac{\sqrt{3}}{1}\right)^2$$

1 mark for  $\sec\left(\frac{5\pi}{4}\right)$  (½ mark for quadrant; ½ mark for value)

$$\left(\frac{4}{2}\right)\left(\frac{3}{1}\right)$$

1 mark for  $\tan\left(-\frac{2\pi}{3}\right)$  (½ mark for quadrant; ½ mark for value)

6

**2 marks**

### Exemplar 1

---

$$\left(\frac{2}{\sqrt{2}}\right)^2 \cdot -\sqrt{3}$$

$$\frac{4}{2}$$

$$2 \cdot -\sqrt{3}$$

$$-2\sqrt{3}$$

---

1 out of 2

+ ½ mark for value of  $\sec\left(\frac{5\pi}{4}\right)$

+ ½ mark for value of  $\tan\left(-\frac{2\pi}{3}\right)$

E7 (transcription error in line 1)

### Exemplar 2

---

$$\cos^2\left(\frac{2}{\sqrt{2}}\right) \cdot \tan^2\left(\frac{\sqrt{3}}{1}\right)$$

$$\left(\frac{2}{\sqrt{2}}\right)^2 \cdot \left(\frac{\sqrt{3}}{1}\right)^2$$

$$\frac{4}{2} \cdot \frac{3}{1} = \frac{12}{2} = \boxed{6}$$

---

1 out of 2

+ ½ mark for value of  $\sec\left(\frac{5\pi}{4}\right)$

+ 1 mark for  $\tan\left(-\frac{2\pi}{3}\right)$

– ½ mark for procedural error in line 1

## Question 29

R1

Given the functions  $f(x) = x - 3$  and  $g(x) = \sqrt{x} + 2$ , state the domain of  $g(f(x))$ .

**Solution**

$$\{x \mid x \geq 3, x \in \mathbb{R}\}$$

1 mark

or

$$[3, \infty)$$

### Exemplar 1

---

$$g(x-3) = \sqrt{x-3} + 2$$

$$d: \{x \mid x > 3, x \in \mathbb{R}\}$$

---

1 out of 1

award full marks

E8 (bracket error when stating domain)

### Exemplar 2

---

$$\sqrt{x-3} + 2$$

$$d: (3, \infty)$$

---

1 out of 1

award full marks

E8 (bracket error when stating domain)

## Question 30

T6

Evaluate.

$$\sin 70^\circ \cos 25^\circ - \cos 70^\circ \sin 25^\circ$$

### Solution

$$\sin 70^\circ \cos 25^\circ - \cos 70^\circ \sin 25^\circ = \sin(70^\circ - 25^\circ) \quad \frac{1}{2} \text{ mark for substitution into correct identity}$$

$$= \sin 45^\circ$$

$$= \frac{\sqrt{2}}{2}$$

$\frac{1}{2}$  mark for exact value

**1 mark**

### Exemplar 1

---

$$\sin(70-25)$$

---

½ out of 1

+ ½ mark for substitution into correct identity  
E5 (units of measure omitted in final answer)

## Question 31

R10

Solve, algebraically.

$$\log_{15}(x^2 - 1) = 1$$

### Solution

#### Method 1

$$15^1 = x^2 - 1$$

$$0 = x^2 - 16$$

$$0 = (x + 4)(x - 4)$$

$$x = -4, x = 4$$

1 mark for exponential form

1 mark for solving for  $x$  ( $\frac{1}{2}$  mark for each value)

**2 marks**

#### Method 2

$$\log_{15}(x^2 - 1) = \log_{15} 15$$

$$x^2 - 1 = 15$$

$$x^2 = 16$$

$$x = \pm 4$$

$\frac{1}{2}$  mark for logarithmic form

$\frac{1}{2}$  mark for equating arguments

1 mark for solving for  $x$  ( $\frac{1}{2}$  mark for each value)

**2 marks**

### Exemplar 1

---

$$\begin{aligned}\log_{15} (x^2-1) &= \log_{15} 15 \\ x^2-1 &= 15 \\ \sqrt{x^2} &= \sqrt{16} \\ x &= \pm 4\end{aligned}$$

---

1½ out of 2

award full marks

– ½ mark for procedural error in line 1

### Exemplar 2

---

$$\begin{aligned}15 &= x^2 - 1 \\ x^2 - 1 - 15 &= 0 \\ x^2 - 16 &= 0 \\ (x+4)(x-4) &= 0 \\ \cancel{x=-4} \quad x &= 4\end{aligned}$$

---

1½ out of 2

award full marks

– ½ mark for procedural error (incorrectly rejecting correct answer)

### Exemplar 3

---

$$\begin{aligned}15 &= x^2 - 1 \\ 16 &= x^2 \\ x &= \sqrt{16} \\ \boxed{x=4}\end{aligned}$$

---

1½ out of 2

+ 1 mark for exponential form

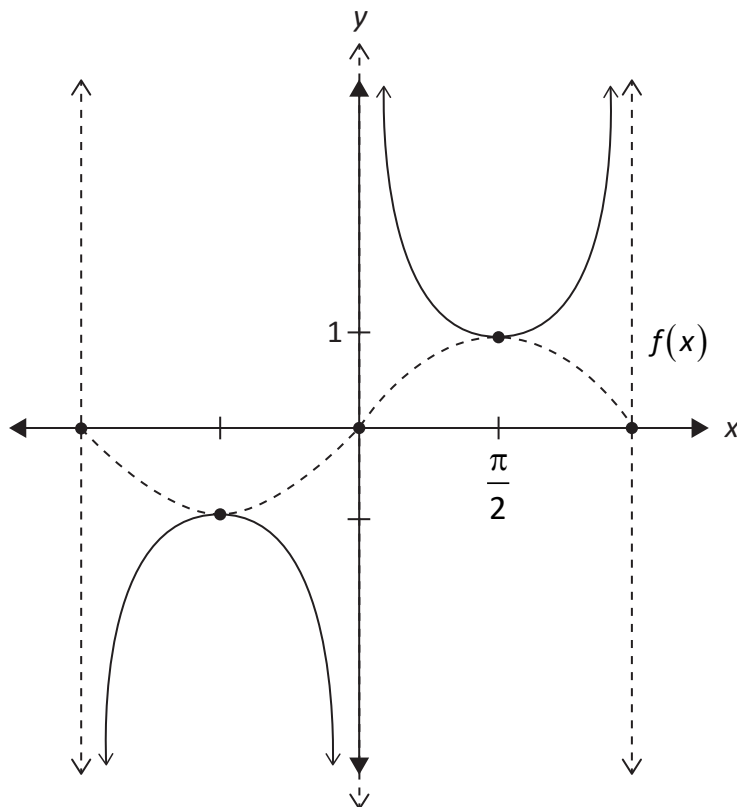
+ ½ mark for solving for x

### Question 32

R1

Given the graph of  $y = f(x)$ , sketch the graph of  $y = \frac{1}{f(x)}$ .

#### Solution



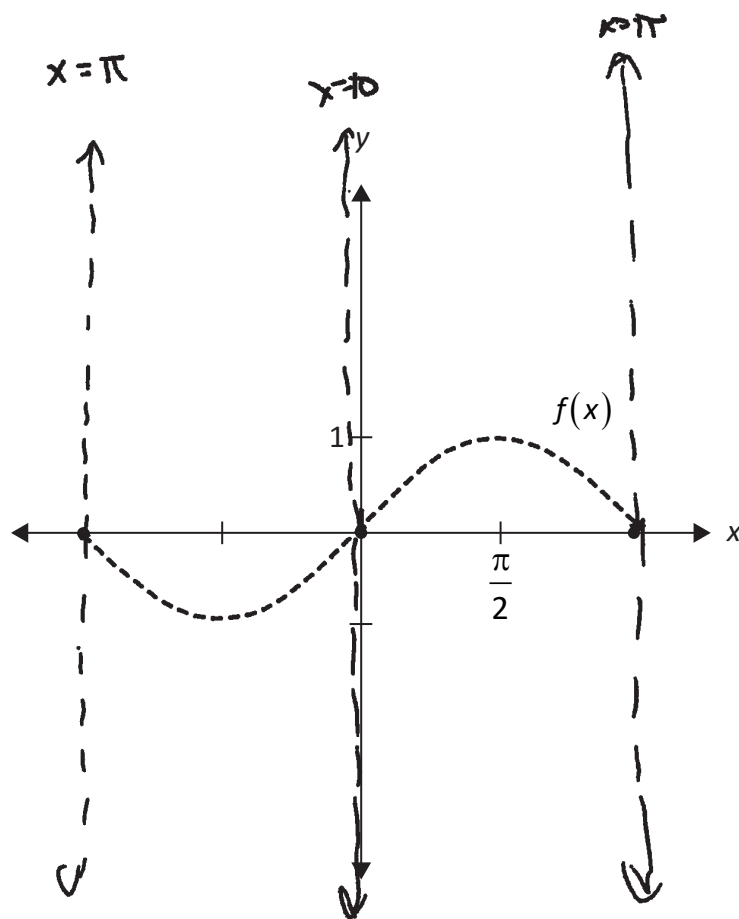
1 mark for asymptotic behaviour approaching  $x = -\pi$ ,  $x = 0$ , and  $x = \pi$

1 mark for invariant points ( $\frac{1}{2}$  mark for  $y = -1$ ;  $\frac{1}{2}$  mark for  $y = 1$ )

**2 marks**

## Exemplar 1

---

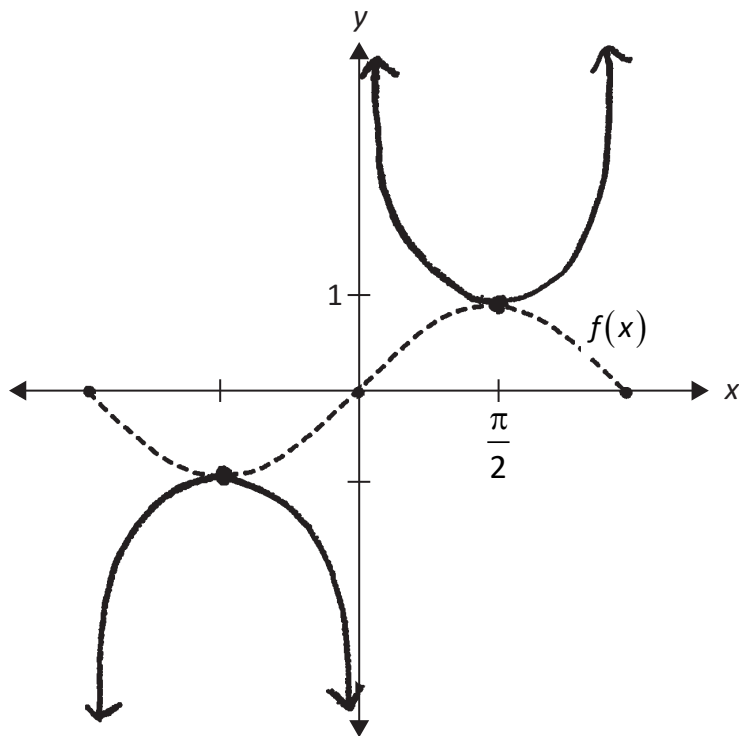


---

0 out of 2

## Exemplar 2

---



---

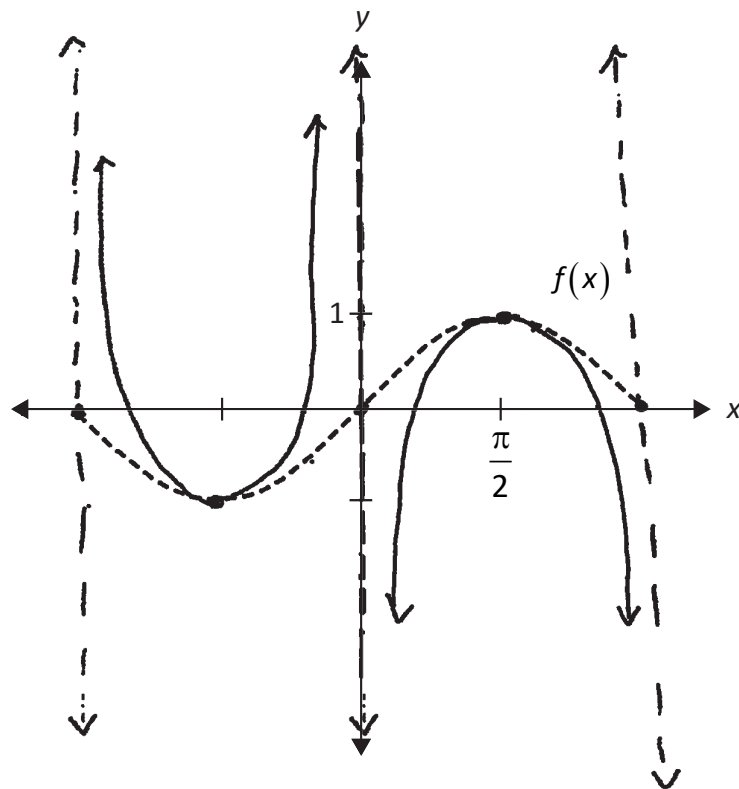
**2 out of 2**

award full marks

E10 (asymptotes omitted but still implied)

### Exemplar 3

---



---

**1 out of 2**

award full marks

– 1 mark concept error (incorrect reciprocal values)

### Question 33

T4

A light is attached to the end of a blade of a wind turbine. The height of the light, with respect to the ground, follows the sinusoidal equation:

$$h(t) = 80 \cos\left(\frac{\pi}{2}t\right) + 240$$

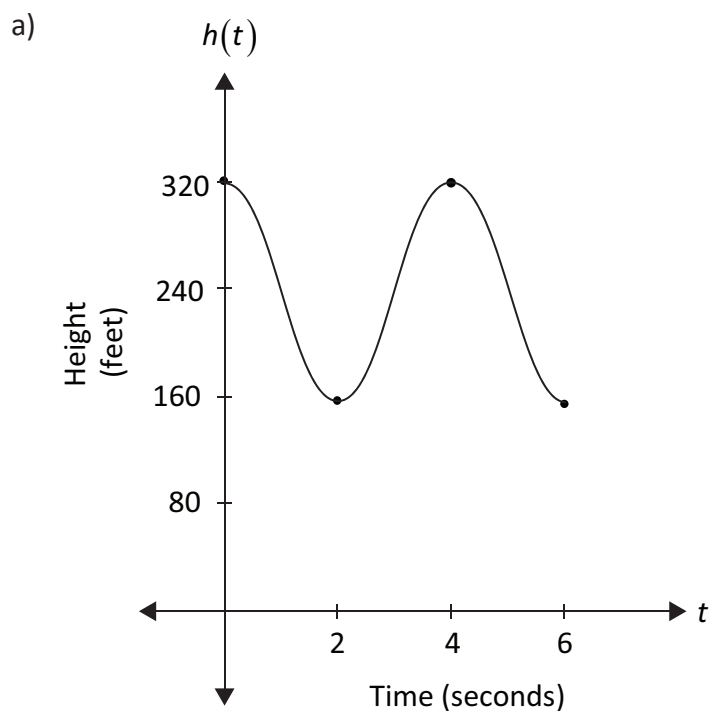
where  $h(t)$  is the height of the light with respect to the ground, measured in feet and,

$t$  is the time, measured in seconds.



- a) Sketch the graph of the height of the light with respect to the ground, over an interval of 6 seconds.
- b) If the turbine begins to rotate faster, describe the resulting effect on the period of the graph of  $h(t)$ .

### Solution



1 mark for shape of a sinusoidal function with correct amplitude  
1 mark for period  
1 mark for vertical translation

**3 marks**

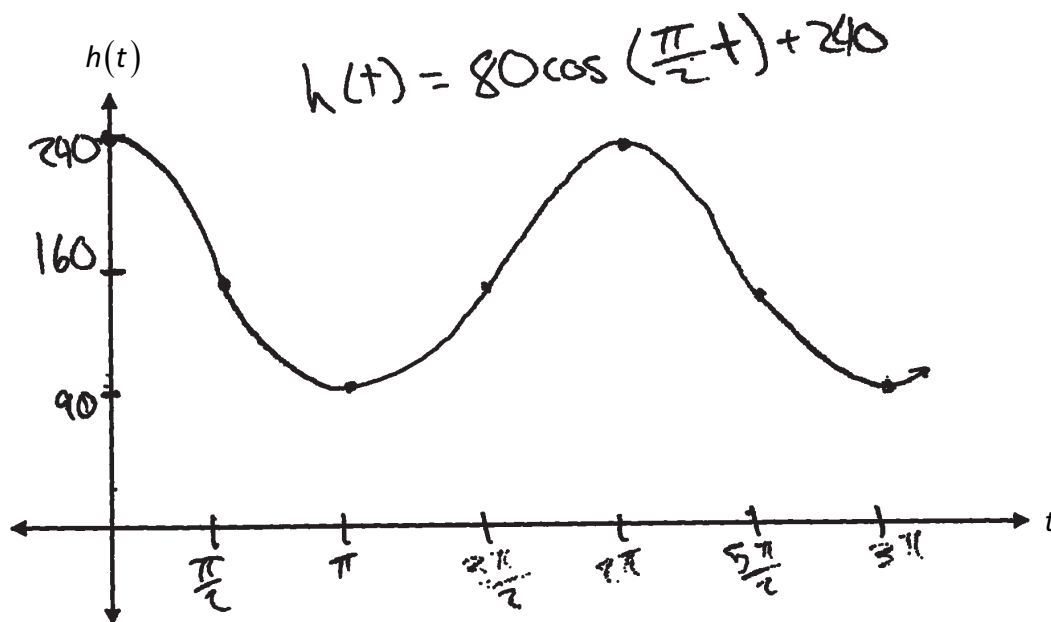
**1 mark**

- b) The period would decrease.

## Exemplar 1

---

a)



---

½ out of 3

+ 1 mark for shape of a sinusoidal function with correct amplitude

– ½ mark for arithmetic error when calculating the amplitude

b)

If the turbine begins to rotate faster, the period of the graph will shorten, and the graph will begin to move up + down faster, or over less seconds.

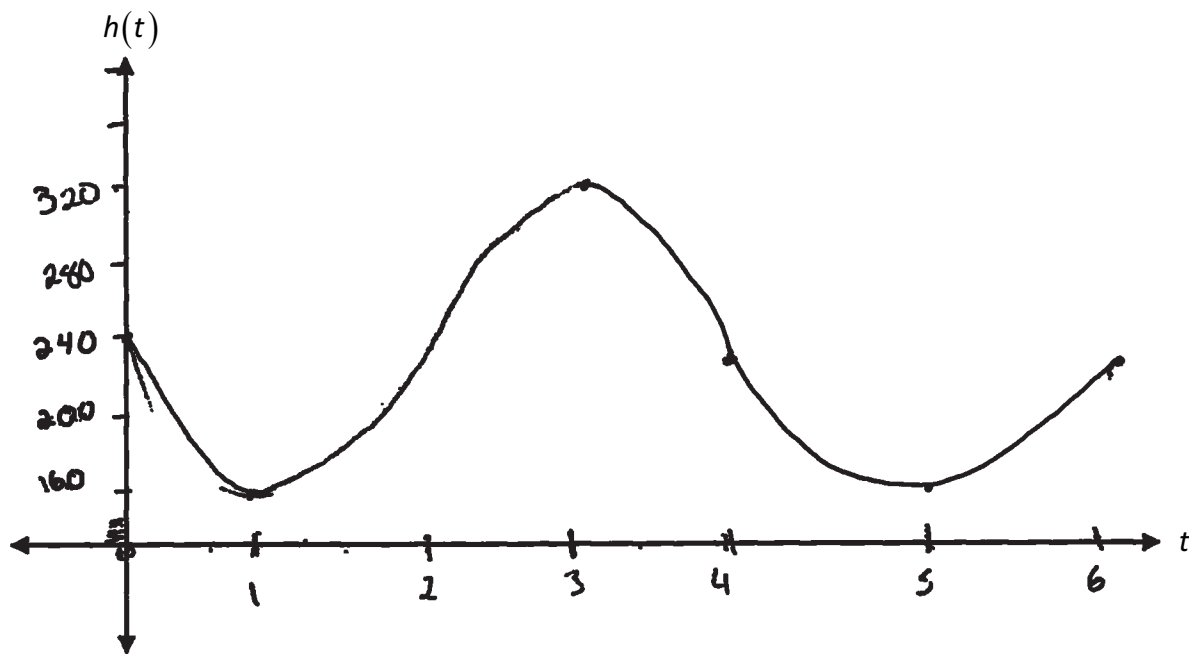
---

1 out of 1

## Exemplar 2

---

a)



---

2 out of 3

award full marks

– 1 mark for concept error (sketching  $y = -\sin t$  instead of  $y = \cos t$ )

b)

The period will become smaller

---

1 out of 1

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## Question 34

P2

Elodie was asked to determine the total number of arrangements of the letters in the word EXCELLENCE.

Her solution:

$$10! = 3628\ 800$$

Describe her error.

### Solution

She did not consider the repeating letters.

1 mark

### Exemplar 1

---

There are letters which repeat that she counted as all separate.  
The proper answer is 5!

---

0 out of 1

### Exemplar 2

---

She didn't factor in that there are letter repeats in the word. The correct solution is  $\frac{10!}{4!2!}$

---

½ out of 1

award full marks

– ½ mark for lack of clarity in explanation

## Question 35

T6

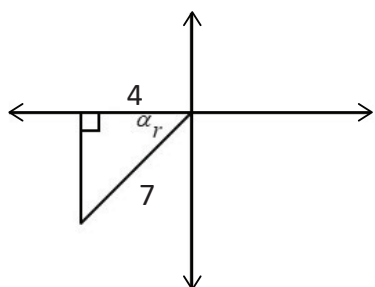
Given that  $\cos \alpha = -\frac{4}{7}$  where  $\alpha$  is in quadrant III and  $\sin \beta = \frac{5}{13}$  where  $\beta$  is in quadrant II, determine the exact value of:

a)  $\cos(\alpha + \beta)$

b)  $\sec(\alpha + \beta)$

### Solution

a)

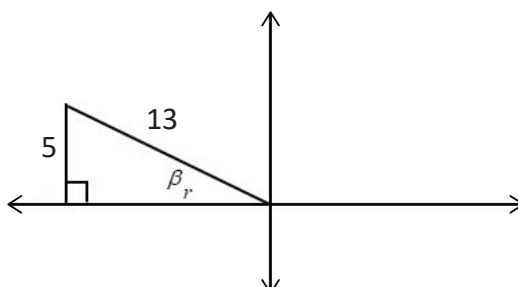


$$x^2 + y^2 = r^2$$

$$16 + y^2 = 49$$

$$y^2 = 33$$

$$y = \pm\sqrt{33}$$



$$x^2 + y^2 = r^2$$

$$x^2 + 25 = 169$$

$$x^2 = 144$$

$$x = \pm 12$$

$\frac{1}{2}$  mark for value of  $x$

$\frac{1}{2}$  mark for value of  $y$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$= \left(-\frac{4}{7}\right)\left(-\frac{12}{13}\right) - \left(-\frac{\sqrt{33}}{7}\right)\left(\frac{5}{13}\right)$$

$$= \frac{48 + 5\sqrt{33}}{91}$$

$\frac{1}{2}$  mark for consistent value of  $\cos \beta$

$\frac{1}{2}$  mark for consistent value of  $\sin \alpha$

1 mark for substitution into correct identity

**3 marks**

b)  $\sec(\alpha + \beta) = \frac{91}{48 + 5\sqrt{33}}$

1 mark for consistent value of  $\sec(\alpha + \beta)$

**1 mark**

### Note:

Accept any of the following values for  $x$ :  $x = \pm 12$ ,  $x = -12$ , or  $x = 12$ .

Accept any of the following values for  $y$ :  $y = \pm\sqrt{33}$ ,  $y = -\sqrt{33}$ , or  $y = \sqrt{33}$ .

### Exemplar 1

---

a)

$$\sin \beta = \frac{5}{13} \quad \cos \alpha = -\frac{4}{7}$$
$$\cos \beta = -\frac{12}{13} \quad \sin \alpha = -\frac{\sqrt{33}}{7}$$

$$\begin{aligned}\cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ &= \left(-\frac{4}{7}\right)\left(-\frac{12}{13}\right) - \left(-\frac{\sqrt{33}}{7}\right)\left(\frac{5}{13}\right) \\ &= \frac{-48}{91} - \frac{5\sqrt{33}}{91} \\ &= \frac{-48 - 5\sqrt{33}}{91}\end{aligned}$$

---

2½ out of 3

award full marks

– ½ mark for arithmetic errors in line 5

b)

$$\begin{aligned}\sec(\alpha + \beta) &= \frac{1}{\cos(\alpha + \beta)} \\ &= \frac{91}{(-48 - 5\sqrt{33})}\end{aligned}$$

---

1 out of 1

## Exemplar 2

---

a)

$$= \cos \left( -\frac{4}{7} + -\frac{12}{13} \right)$$

$$= \cos \left( \frac{-52 + -84}{91} \right)$$

$$= \cos \left( \frac{-136}{91} \right)$$

$$= \cos(-45)$$

$$= -\frac{\sqrt{2}}{2}$$

---

½ out of 3

+ ½ mark for value of x

b)

$$= \frac{-2}{\sqrt{2}}$$

---

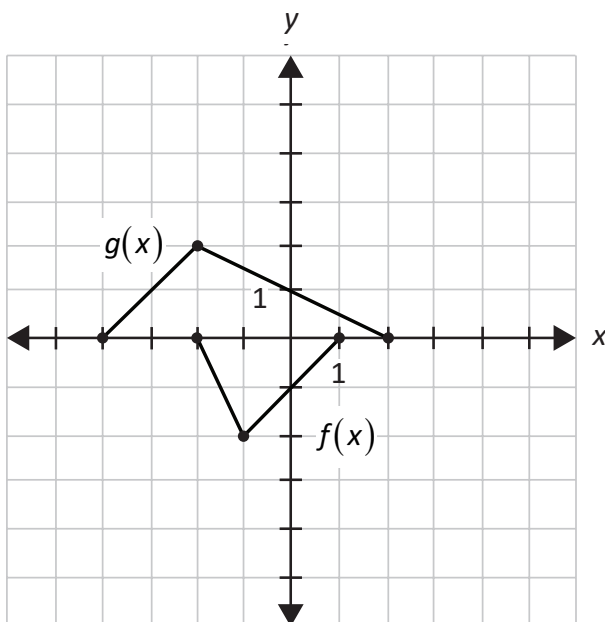
1 out of 1

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## Question 36

R4, R5

Determine the equation for  $g(x)$ , in terms of  $f(x)$ .



**Solution**

$$g(x) = -f\left(\frac{x}{2}\right)$$

1 mark for vertical reflection  
1 mark for horizontal stretch

**2 marks**

### Exemplar 1

---

$$g(x) = \underline{\quad -f(2x) \quad}$$

---

1 out of 2

+ 1 mark for vertical reflection

### Exemplar 2

---

$$g(x) = \underline{\quad f\left(\frac{1}{2}x\right) \quad}$$

---

1 out of 2

+ 1 mark for horizontal stretch

### Exemplar 3

---

$$g(x) = \underline{\quad -g\left(\frac{1}{2}x\right) \quad}$$

---

1½ out of 2

award full marks

– ½ mark for procedural error ( $g$  instead of  $f$ )

### Exemplar 4

---

$$g(x) = \underline{\quad \left(-\frac{1}{2}x\right) \quad}$$

---

1 out of 2

award full marks

– 1 mark for concept error (not writing an equation in term of  $f(x)$ )

## Question 37

R7

State possible values for  $m$  and  $n$  that satisfy the equation,  $\log_m n = 5$ .

### Solution

Some possible solutions are:

$$m = 2 \quad n = 32$$

or

$$m = 3 \quad n = 243$$

or

$$m = 4 \quad n = 1024$$

or

$$m = 5 \quad n = 3125$$

1 mark

### Notes:

Other values for  $m$  and  $n$  are possible.

### Exemplar 1

---

$$\log_2 32 = 5$$

---

1 out of 1

### Exemplar 2

---

$$2^5 = 32$$

---

1 out of 1

award full marks

E1 (final answer not stated)

### Exemplar 3

---

$$m = 1$$
$$n = 1$$

---

0 out of 1

## Question 38

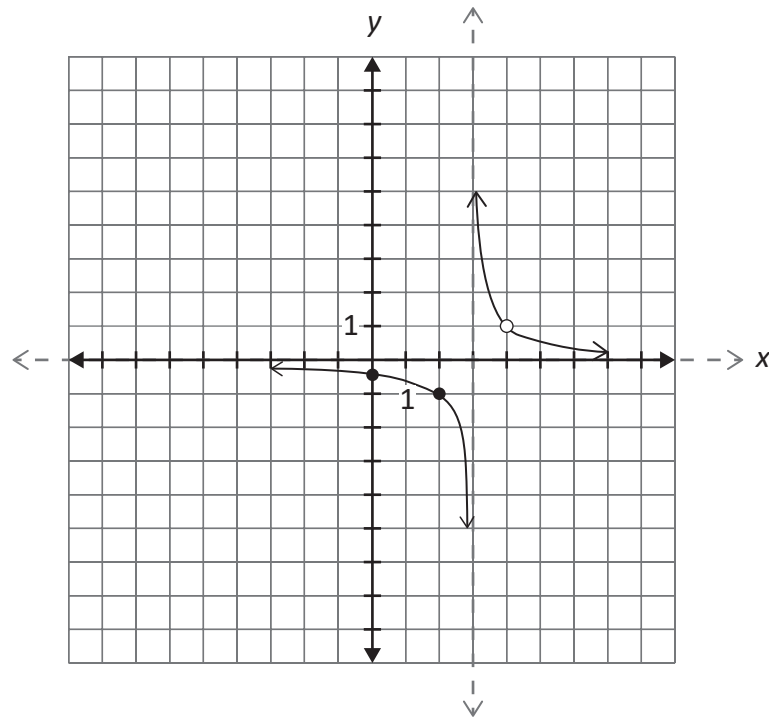
R14

Sketch the graph of  $f(x) = \frac{x-4}{x^2-7x+12}$ .

### Solution

$$f(x) = \frac{\cancel{(x-4)}}{\cancel{(x-4)}(x-3)} \\ = \frac{1}{x-3}, \quad x \neq 4$$

$\therefore$  there is a point of discontinuity (hole) at (4,1)



1 mark for asymptotic behaviour approaching  $x = 3$

1 mark for asymptotic behaviour approaching  $y = 0$

1 mark for point of discontinuity (hole) at  $x = 4$

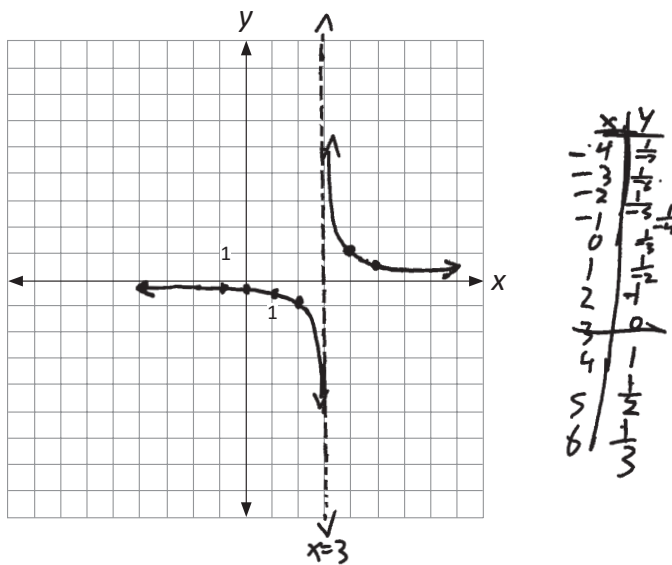
1 mark for shape of rational function ( $\frac{1}{2}$  mark for graph left of  $x = 3$ ,  $\frac{1}{2}$  mark for graph right of  $x = 3$ )

**4 marks**

### Notes:

Deduct  $\frac{1}{2}$  mark for procedural error (incorrect  $y$ -value for point of discontinuity (hole)).

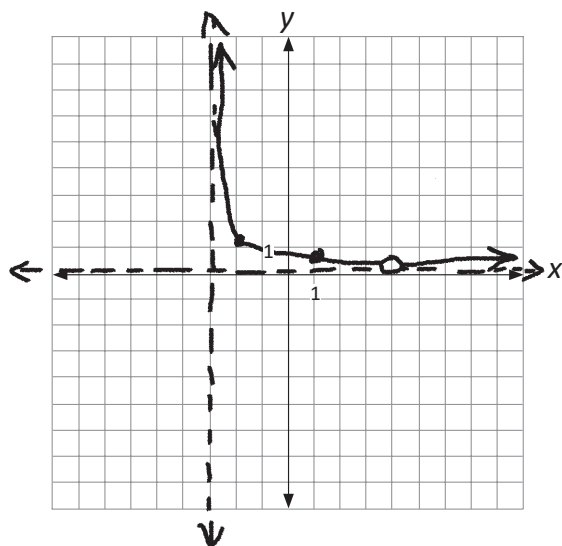
## Exemplar 1



3 out of 4

- + 1 mark for asymptotic behaviour approaching  $x = 3$
  - + 1 mark for asymptotic behaviour approaching  $y = 0$
  - + 1 mark for shape of rational function
- E10 (asymptote omitted but still implied)

## Exemplar 2



$$f(x) = \frac{x-4}{(x-4)(x+3)}$$

$$f(x) = \frac{1}{x+3}$$

3 out of 4

- + 1 mark for asymptotic behaviour approaching consistent vertical asymptote
- + 1 mark for asymptotic behaviour approaching  $y = 0$
- + 1 mark for point of discontinuity (hole) at  $x = 4$
- + ½ mark for consistent shape of graph right of the vertical asymptote
- ½ mark for arithmetic error (incorrect factoring)

## Question 39

P2

Justify that the value of  ${}_3P_4$  does not exist.

### Solution

#### Method 1

When using  ${}_nP_r$ ,  $n \geq r$ .

#### Method 2

$${}_3P_4 = \frac{3!}{(3-4)!}$$

$${}_3P_4 = \frac{3!}{(-1)!}$$

$(-1)!$  does not exist, therefore  ${}_3P_4$  does not exist.

1 mark

### Exemplar 1

---

Selection must be less than or equal to the total.

---

1 out of 1

### Exemplar 2

---

In  $n^Pr$ ,  $n$  must be greater than  $r$ .

---

½ out of 1

award full marks

– ½ mark for lack of clarity in explanation

## Question 40

T4

State the range of the sinusoidal function that has the following characteristics:

- an amplitude of 3
- a maximum at  $\left(\frac{\pi}{2}, 1\right)$

### Solution

$$\{y \mid -5 \leq y \leq 1, y \in \mathbb{R}\}$$

or

$$[-5, 1]$$

1 mark

### Exemplar 1

---

Range:  $[1, -5]$

---

**1 out of 1**

award full marks

E8 (range written in incorrect order)

### Exemplar 2

---

Range:  $(-5, 1)$

---

**1 out of 1**

award full marks

E8 (bracket error when stating range)

### Exemplar 3

---

Range:  $[-2, 1]$

---

**0 out of 1**

## Question 41

P3

Solve, algebraically.

$${}_nC_2 = 15$$

### Solution

$$\frac{n!}{2!(n-2)!} = 15$$

½ mark for substitution

$$\frac{n(n-1)\cancel{(n-2)!}}{2!\cancel{(n-2)!}} = 15$$

1 mark for factorial expansion

½ mark for simplification of factorials

$$n(n-1) = 30$$

$$n^2 - n - 30 = 0$$

$$(n-6)(n+5) = 0$$

$$n = 6 \quad \cancel{n = -5}$$

½ mark for the permissible value of  $n$

½ mark for showing the rejection of the extraneous root

**3 marks**

### Exemplar 1

---

$$\frac{n!}{2!(n-2)!} = 0$$

$$\frac{(n)(n-1)(\cancel{n-2})!}{2(\cancel{n-2})!} = 0$$

$$\frac{(n)(n-1)}{2} = 0$$

$$\frac{n(n-1)}{n} = \frac{0}{n}$$

$$n-1 = 0$$

$$\boxed{n=1}$$

---

2 out of 3

- + ½ mark for substitution
  - + 1 mark for factorial expansion
  - + ½ mark for simplification of factorials
- E7 (transcription error in line 1)

## Exemplar 2

---

$$\frac{n!}{2!(n-2)!} = 15$$

$$\frac{(n)(n-1)(\cancel{n-2})!}{2! \cdot 2!(\cancel{n-2})!} = 15 \cdot 2!$$

$$n(n-1) = 30$$

$$\begin{aligned} 6(6-1) &= 30 \\ 6(5) &= 30 \\ n &= 6 \end{aligned}$$

---

**2½ out of 3**

- + ½ mark for substitution
  - + 1 mark for factorial expansion
  - + ½ mark for simplification of factorials
  - + ½ mark for the permissible value of  $n$
- E7 (notation error in line 2)

### Exemplar 3

---

$$nC_2 = \frac{n!}{2!(n-2)!} = 15$$

$$nC_2 = \frac{(n-1) \cancel{2!}}{2! \cancel{2!}} = 15$$

$$\frac{(n-1)}{2} = 15$$

$$\frac{(n-1) \cdot 2}{2} = 15 \cdot 2$$

$$n-1 = 30$$

$$n = 31$$

---

1½ out of 3

+ ½ mark for substitution

+ ½ mark for simplification of factorials

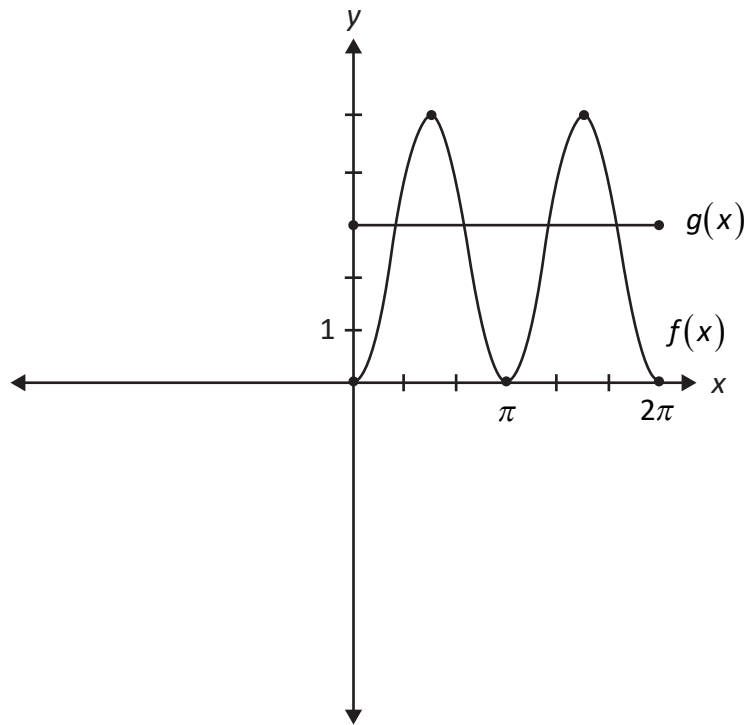
+ ½ mark for the permissible value of  $n$

E4 (missing brackets but still implied in line 2)

## Question 42

T5

Describe how to use the graphs of  $f(x) = 5\sin^2 x$  and  $g(x) = 3$  to solve the equation,  $5\sin^2 x = 3$ , over the interval  $[0, 2\pi]$ .



### Solution

The solution will be the  $x$ -values where the two graphs intersect.

1 mark

### Exemplar 1

---

The points where the  
two graphs intersect.  
are the solutions to  
 $5\sin^2 x = 3$ .

---

½ out of 1

award full marks

– ½ mark for lack of clarity in explanation

### Exemplar 2

---

The points of  
intersection will  
be the solution.

---

½ out of 1

award full marks

– ½ mark for lack of clarity in explanation

## Question 43

R8

Solve, algebraically.

$$2\log_3 5 - \frac{1}{3}\log_3 125 = \log_3 a$$

### Solution

#### Method 1

$$\log_3 5^2 - \log_3 125^{\frac{1}{3}} = \log_3 a$$

1 mark for power law ( $\frac{1}{2}$  mark for each)

$$\log_3 25 - \log_3 5 = \log_3 a$$

$$\log_3 \left( \frac{25}{5} \right) = \log_3 a$$

1 mark for quotient law

$$\log_3 (5) = \log_3 a$$

$$a = 5$$

1 mark for equating arguments

**3 marks**

#### Method 2

$$\log_3 5^2 - \log_3 125^{\frac{1}{3}} - \log_3 a = 0$$

1 mark for power law ( $\frac{1}{2}$  mark for each)

$$\log_3 \left( \frac{25}{5a} \right) = 0$$

1 mark for quotient law

$$3^0 = \frac{25}{5a}$$

1 mark for exponential form

$$5a = 25$$

$$a = 5$$

**3 marks**

### Exemplar 1

---

$$\log_3 25 - \log_3 \frac{1}{5} = \log_3 a$$

$$\log_3 5 = \log_3 a$$

$$5 = a$$

---

1½ out of 3

+ ½ mark for power law

+ 1 mark for equating arguments

### Exemplar 2

---

$$\log_3 5^2 - \log_3 125^{\frac{1}{3}} = \log_3 a$$

$$\cancel{\log_3} \left( \frac{5^2}{\sqrt[3]{125}} \right) = \cancel{\log_3} a$$

$$\frac{25}{5} = a$$

$$a = 5$$

---

2½ out of 3

award full marks

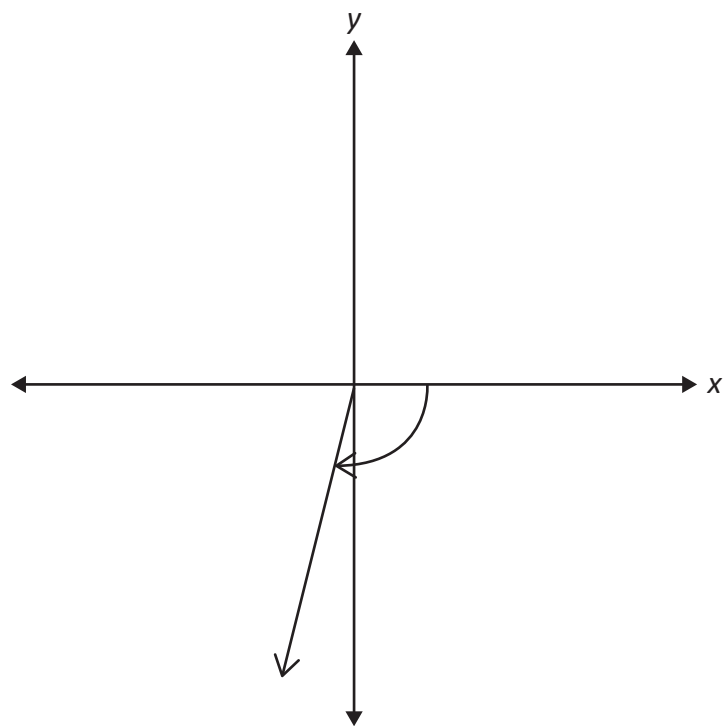
– ½ mark for procedural error (crossing out the logarithms)

## Question 44

T1

Sketch the angle,  $\frac{-7\pi}{12}$ , in standard position.

**Solution**

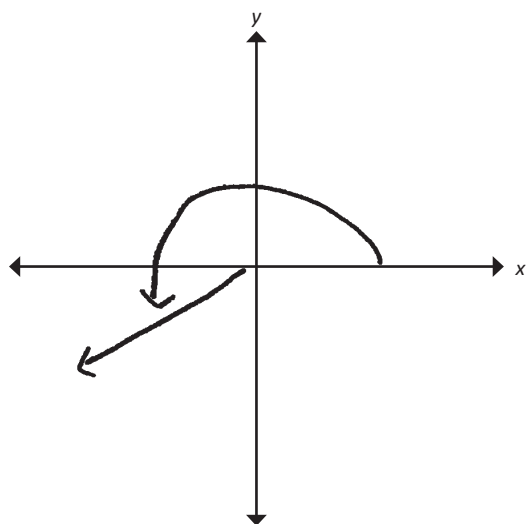


½ mark for an appropriate angle  
in quadrant III  
½ mark for correct direction

**1 mark**

### Exemplar 1

---

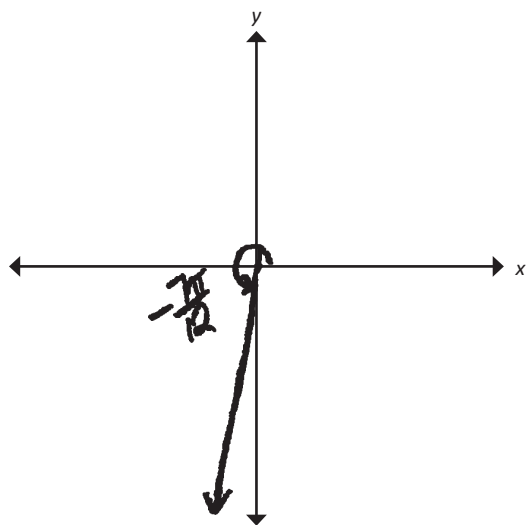


---

0 out of 1

### Exemplar 2

---



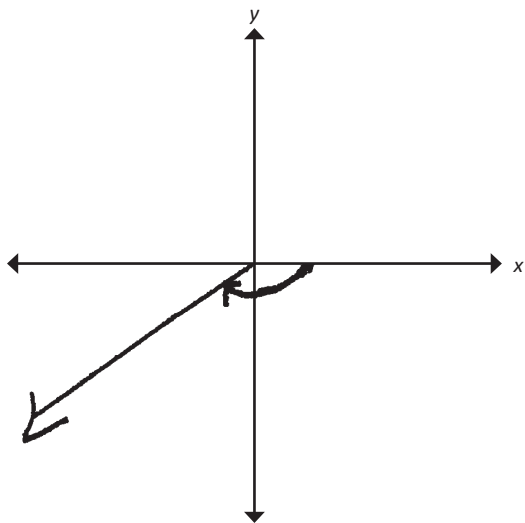
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$\frac{1}{2}$  out of 1

+  $\frac{1}{2}$  mark for an appropriate angle in quadrant III

### Exemplar 3

---



---

**$\frac{1}{2}$  out of 1**

+  $\frac{1}{2}$  mark for correct direction

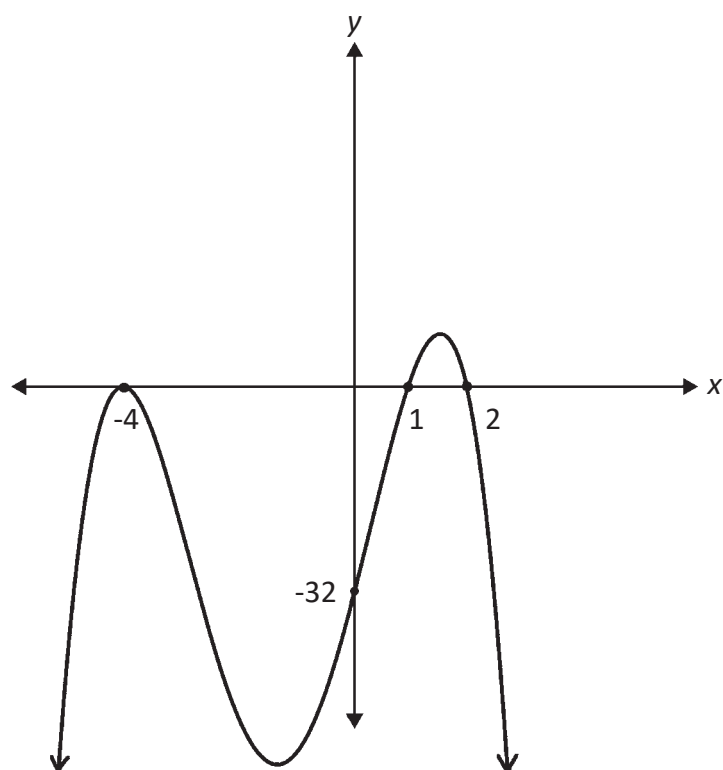
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## Question 45

R12

Sketch the graph of  $p(x) = -(x-1)(x-2)(x+4)^2$ .

### Solution

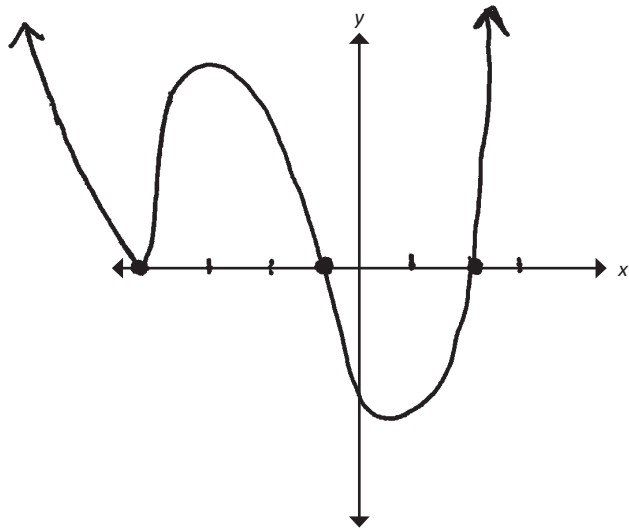


1 mark for x-intercepts  
½ mark for y-intercept  
1 mark for multiplicity of 2 at  $x = -4$   
½ mark for end behaviour

**3 marks**

## Exemplar 1

---



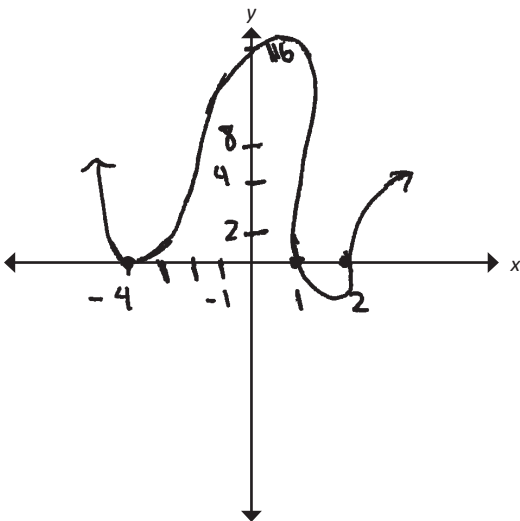
---

**1½ out of 3**

- + 1 mark for x-intercepts
  - + 1 mark for multiplicity of 2 at  $x = -4$
  - ½ mark for procedural error (one incorrect x-intercept)
- E9 (scale values on axes not indicated)

## Exemplar 2

---



---

**1½ out of 3**

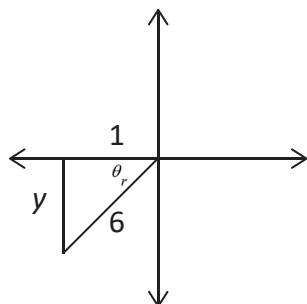
- + 1 mark for x-intercepts
  - + 1 mark for multiplicity of 2 at  $x = -4$
  - ½ mark for incorrect shape of graph
- E9 (scale values on axes incorrectly spaced)

## Question 46

T3

The point,  $P(\theta) = \left(-\frac{1}{6}, y\right)$ , lies on the unit circle and is located in Quadrant III. Determine the exact value of  $\csc \theta$ .

### Solution



$$(-1)^2 + y^2 = (6)^2$$

½ mark for substitution

$$1 + y^2 = 36$$

$$y^2 = 35$$

$$y = \pm\sqrt{35}$$

½ mark for solving for y

$$\csc \theta = -\frac{6}{\sqrt{35}}$$

1 mark for consistent value of  $\csc \theta$   
(½ mark for quadrant; ½ mark for value)

or

$$\csc \theta = -\frac{6\sqrt{35}}{35}$$

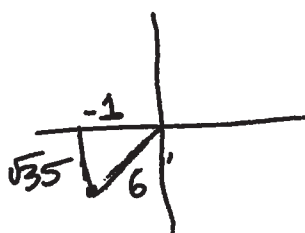
**2 marks**

### Notes:

Accept any of the following values for  $y$  :  $y = \pm\sqrt{35}$ ,  $y = \sqrt{35}$ , or  $y = -\sqrt{35}$ .

### Exemplar 1

---

$$\frac{1}{\sin \theta}$$

$$\csc \theta = \frac{6}{\sqrt{35}}$$

---

1½ out of 2

- + ½ mark for substitution
- + ½ mark for solving for y
- + ½ mark for value of  $\csc \theta$

### Exemplar 2

---

$$\left(-\frac{1}{6}\right)^2 + y^2 = 1^2$$

$$\frac{1}{36} + y^2 = 1$$

$$y^2 = \frac{35}{36}$$

$$y = -\sqrt{\frac{35}{36}}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\csc \theta = \frac{1}{-\sqrt{\frac{35}{36}}}$$

---

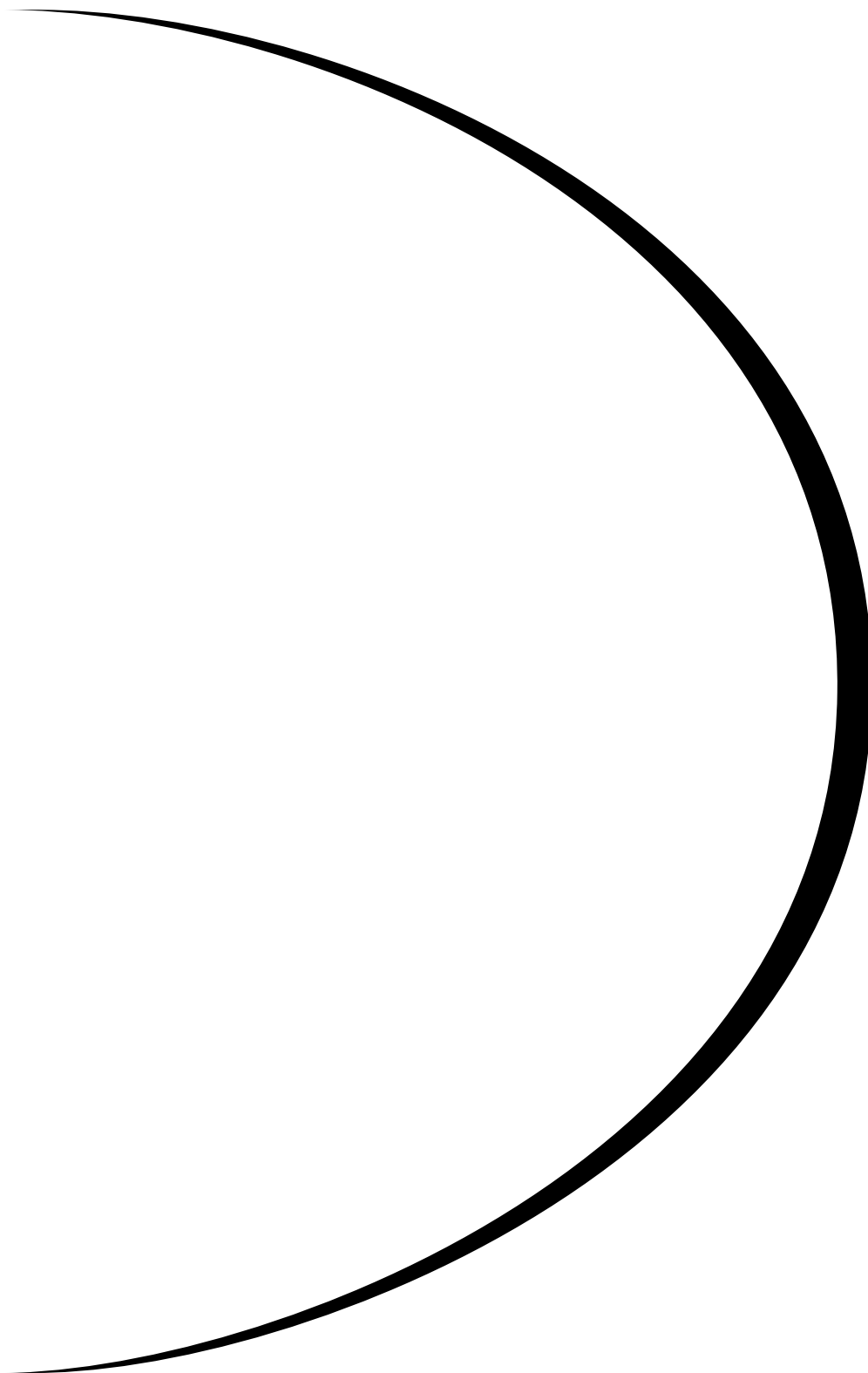
1½ out of 2

award full marks

- ½ mark for procedural error (not taking the square root of both the numerator and denominator)
- E1 (answer given as a complex fraction)

# Appendices

---



# Appendix A

## Marking Guidelines

Errors that are conceptually related to the learning outcomes associated with the question will result in a 1 mark deduction.

Each time a student makes one of the following errors, a  $\frac{1}{2}$  mark deduction will apply:

- arithmetic error
- procedural error
- terminology error in explanation
- lack of clarity in explanation, description, or justification
- incorrect shape of graph (only when marks are not allocated for shape)

### Communication Errors

The following errors, which are not conceptually related to the learning outcomes associated with the question, may result in a  $\frac{1}{2}$  mark deduction and will be tracked on the *Answer/Scoring Sheet*.

|                                      |                                                                                                                                                                                                                             |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>E1<br/>final answer</b>           | <ul style="list-style-type: none"><li>▪ answer given as a complex fraction</li><li>▪ final answer not stated</li><li>▪ impossible solution(s) not rejected in final answer and/or in step leading to final answer</li></ul> |
| <b>E2<br/>equation/expression</b>    | <ul style="list-style-type: none"><li>▪ changing an equation to an expression or vice versa</li><li>▪ equating the two sides when proving an identity</li></ul>                                                             |
| <b>E3<br/>variables</b>              | <ul style="list-style-type: none"><li>▪ variable omitted in an equation or identity</li><li>▪ variables introduced without being defined</li></ul>                                                                          |
| <b>E4<br/>brackets</b>               | <ul style="list-style-type: none"><li>▪ "sin x<sup>2</sup>" written instead of "sin<sup>2</sup> x"</li><li>▪ missing brackets but still implied</li></ul>                                                                   |
| <b>E5<br/>units</b>                  | <ul style="list-style-type: none"><li>▪ units of measure omitted in final answer</li><li>▪ incorrect units of measure</li><li>▪ answer stated in degrees instead of radians or vice versa</li></ul>                         |
| <b>E6<br/>rounding</b>               | <ul style="list-style-type: none"><li>▪ rounding error</li><li>▪ rounding too early</li></ul>                                                                                                                               |
| <b>E7<br/>notation/transcription</b> | <ul style="list-style-type: none"><li>▪ notation error</li><li>▪ transcription error</li></ul>                                                                                                                              |
| <b>E8<br/>domain/range</b>           | <ul style="list-style-type: none"><li>▪ answer outside the given domain</li><li>▪ bracket error made when stating domain or range</li><li>▪ domain or range written in incorrect order</li></ul>                            |
| <b>E9<br/>graphing</b>               | <ul style="list-style-type: none"><li>▪ endpoints or arrowheads omitted or incorrect</li><li>▪ scale values on axes not indicated or incorrectly spaced</li><li>▪ coordinate points labelled incorrectly</li></ul>          |
| <b>E10<br/>asymptotes</b>            | <ul style="list-style-type: none"><li>▪ asymptotes drawn as solid lines</li><li>▪ asymptotes omitted but still implied</li><li>▪ graph crosses or curls away from asymptotes</li></ul>                                      |

# Appendix B

## Irregularities in Provincial Tests

### A Guide for Local Marking

During the marking of provincial tests, irregularities are occasionally encountered in test booklets. The following list provides examples of irregularities for which an *Irregular Test Booklet Report* should be completed and sent to the department:

- completely different penmanship in the same test booklet
- incoherent work with correct answers
- notes from a teacher indicating how he or she has assisted a student during test administration
- student offering that he or she received assistance on a question from a teacher
- student submitting work on unauthorized paper
- evidence of cheating or plagiarism
- disturbing or offensive content
- no responses provided by the student or only incorrect responses (“0”)

Student comments or responses indicating that the student may be at personal risk of being harmed or of harming others are personal safety issues. This type of student response requires an immediate and appropriate follow-up at the school level. In this case, please ensure the department is made aware that follow-up has taken place by completing an *Irregular Test Booklet Report*.

Except in the case of cheating or plagiarism where the result is a provincial test mark of 0%, it is the responsibility of the division or the school to determine how they will proceed with irregularities. Once an irregularity has been confirmed, the marker prepares an *Irregular Test Booklet Report* documenting the situation, the people contacted and the follow-up. The original copy of this report is to be returned by the local jurisdiction and a copy is to be sent to the department along with the test materials.

## Irregular Test Booklet Report

Test: \_\_\_\_\_

Date marked: \_\_\_\_\_

Booklet No.: \_\_\_\_\_

\_\_\_\_\_

Problem(s) noted: \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

Question(s) affected: \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

Action taken or rationale for assigning marks: \_\_\_\_\_

\_\_\_\_\_

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**Follow up:** \_\_\_\_\_

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**Decision:** \_\_\_\_\_

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**Marker's Signature:** \_\_\_\_\_

**Principal's Signature:** \_\_\_\_\_

**For Department Use Only - After Marking Complete**

**Consultant:** \_\_\_\_\_

**Date:** \_\_\_\_\_

# Appendix C

**Table of Questions by Unit and Learning Outcome**

| Unit A: Transformations of Functions |                  |      |
|--------------------------------------|------------------|------|
| Question                             | Learning Outcome | Mark |
| 7                                    | R2, R3, R5       | 4    |
| 10                                   | R6               | 2    |
| 14                                   | R1               | 2    |
| 17                                   | R5               | 1    |
| 29                                   | R1               | 1    |
| 32                                   | R1               | 2    |
| 36                                   | R4, R5           | 2    |
| Unit B: Trigonometric Functions      |                  |      |
| Question                             | Learning Outcome | Mark |
| 5a)                                  | T1               | 1    |
| 5b)                                  | T1               | 1    |
| 22                                   | T4               | 1    |
| 24                                   | T1               | 1    |
| 28                                   | T3               | 2    |
| 33a)                                 | T4               | 3    |
| 33b)                                 | T4               | 1    |
| 40                                   | T4               | 1    |
| 44                                   | T1               | 1    |
| 46                                   | T3               | 2    |
| Unit C: Binomial Theorem             |                  |      |
| Question                             | Learning Outcome | Mark |
| 1                                    | P1               | 1    |
| 3                                    | P4               | 3    |
| 9                                    | P4               | 1    |
| 20                                   | P2               | 1    |
| 34                                   | P2               | 1    |
| 39                                   | P2               | 1    |
| 41                                   | P3               | 3    |
| Unit D: Polynomial Functions         |                  |      |
| Question                             | Learning Outcome | Mark |
| 6                                    | R12              | 1    |
| 16                                   | R11              | 1    |
| 21                                   | R11              | 1    |
| 25                                   | R11              | 3    |
| 45                                   | R12              | 3    |

| Unit E: Trigonometric Equations and Identities |                  |      |
|------------------------------------------------|------------------|------|
| Question                                       | Learning Outcome | Mark |
| 4                                              | T5               | 3    |
| 12                                             | T6               | 3    |
| 26                                             | T5               | 3    |
| 30                                             | T6               | 1    |
| 35a)                                           | T6               | 3    |
| 35b)                                           | T6               | 1    |
| 42                                             | T5               | 1    |
| Unit F: Exponents and Logarithms               |                  |      |
| Question                                       | Learning Outcome | Mark |
| 2                                              | R10              | 2    |
| 13                                             | R9               | 1    |
| 18                                             | R7               | 1    |
| 23                                             | R8               | 1    |
| 27a)                                           | R9               | 1    |
| 27b)                                           | R9               | 2    |
| 31                                             | R10              | 2    |
| 37                                             | R7               | 1    |
| 43                                             | R8               | 3    |
| Unit G: Radicals and Rationals                 |                  |      |
| Question                                       | Learning Outcome | Mark |
| 8                                              | R14              | 2    |
| 11                                             | R13              | 2    |
| 15                                             | R13              | 4    |
| 19                                             | R14              | 1    |
| 38                                             | R14              | 4    |